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The EUV Snapshot Imaging Spectrograph

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ABSTRACT

The EUV Snapshot Imaging Spectrograph (ESIS) is a next generation rocket borne instrument designed to investigate magnetic reconnection and energy transport in the solar atmosphere by observing emission lines formed in the chromosphere (He I 584.334 Å), the transition region (O V 629.732 Å), and corona (Mg X 609.793 Å). The instrument is a pseudo Gregorian telescope with an octagonal field stop at prime focus. This field stop is re-imaged using an array of four spherical diffraction gratings with differing dispersion angles oriented in 45° increments, with each diffraction grating projecting the spectrum onto a unique detector. The instrument is designed to be capable of obtaining co-temporal spatial ($0.75''\,\mathrm{pix}^{-1}$) and spectral ($36.7\ \mathrm{m\AA\,pix}^{-1}$) images at high cadence ($\geq$1.2 s). Combining the co-temporal exposures from all the detectors enables us to reconstruct line profile information at high spatial and spectral resolution over a large ($11.5'$) field of view. The instrument was launched on 2019 September 30.

1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as: $I(x, y, \lambda, t)$, where x and y are the helioprojective Cartesian coordinates (W. T. Thompson 2006), λ is wavelength, and t is time. The ideal solar imaging spectrograph would capture $I(x, y, \lambda, t)$ with high resolution in x, y, λ, and t *and* over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time t_0: $I(x, y, \lambda, t_0)$. Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than ~150 nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as the Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker lines within them bound the velocity resolution near $\sim$1000 km s^{-1} (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records λ against y without ambiguity, and x is recovered by stepping the slit across the target. The Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and the Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity:

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neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along x. The Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the Solar eruptioN Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024), flown in 2025, demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once, so every emission line forms an image of the Sun and those images superimpose on the detector. The resulting frame, an *overlappogram*, encodes x and λ along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as spaceborne solar spectroscopy, having been recorded by the Naval Research Laboratory's (NRL's) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by the Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

The Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed $m = 0$ order and the $m = \pm 1$ orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a plane, they fill the cylindrical volume of a sounding rocket payload poorly.

The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary aperture stop, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section 3 states the scientific objectives and the requirements they impose. Section 4 describes the instrument itself, Section 5 the mission profile, and Section 6 looks ahead to the next flight of ESIS and to an orbital instrument.

2. THE ESIS CONCEPT

A primary goal of the ESIS instrument is to improve upon the imaging spectroscopy demonstrated by MOSES. Therefore, the design of the new instru-

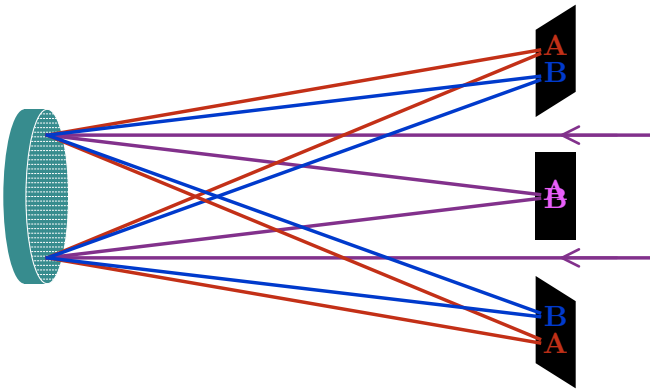


Figure 1. Schematic diagram of the MOSES instrument. Incident light from the right forms an undispersed image on the central $m = 0$ CCD. Dispersed images are formed on the outboard $m = \pm 1$ CCDs. The imaging of longer and shorter wavelengths is indicated by the use of red and blue letters respectively.

ment draws heavily from experiences and lessons learned through two flights of MOSES. ESIS and MOSES are both CTIS instruments. As such, both produce overlappograms of a narrow portion of the solar spectrum, with the goal of enabling the reconstruction of a spectral line profile at every point in the field of view. The similarities end there, however, as the optical layout of ESIS differs significantly from that of MOSES. In this section, we detail some difficulties and limitations encountered with MOSES, then describe how the new design of ESIS addresses these issues.

2.1. *Limitations of the MOSES Design*

The MOSES design features a single concave diffraction grating forming images on three charge-coupled device (CCD) detectors (J. L. Fox et al. 2010) (Figure 1). The optical path is folded in half by a single flat secondary mirror (omitted in Figure 1). Provided that the three cameras are positioned correctly, this arrangement allows the entire telescope to be brought into focus using only the central (undispersed) order and a visible light source. Unfortunately this design uses volume inefficiently for two reasons. First, the lack of magnification by the secondary mirror limits the folded length of the entire telescope to be no less than half of the 5 m focal length of the grating (J. L. Fox et al. 2010; J. L. Fox 2011). Second, the resolving power of the instrument is limited by the ratio of camera separation to pixel size. To achieve the maximum dispersion of 29 $\mathrm{km\,s^{-1}\,pix^{-1}}$ (J. L. Fox et al. 2010), the outboard orders are imaged as far apart as possible in the ~ 0.56 m diameter envelope of the rocket payload. This planar arrangement leaves much unused space in the cylindrical volume of the payload.

Furthermore, the monolithic secondary, though it confers the focus advantage noted above, does not allow efficient placement of the $m = \pm 1$ CCDs. For all practical purposes, the payload can only accommodate three diffraction orders ($m = -1, 0, +1$). Therefore, *MOSES can only collect, at most, three pieces of information at each point in the field of view.* From this, it is not reasonable to expect the reconstruction of more than three degrees of freedom for each spectral line, except in the case of very compact, isolated features such as those described by J. L. Fox et al. (2010) and T. Rust & C. C. Kankelborg (2019). Consequently, it is a reasonable approximation to say that MOSES is sensitive primarily to spectral line intensities, shifts, and widths (C. C. Kankelborg & R. J. Thomas 2001). With any tomographic apparatus, the degree of detail that can be resolved in the object depends critically on the number of viewing angles (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008). So it is with the spectrum we observe with MOSES: more dispersed images are required to confer sensitivity to finer spectral details such as higher moments of the spectral line shape.

A related issue stems from the use of a single dispersion plane. Since the solar corona and transition region are structured by magnetic fields, the scene tends to be dominated by field-aligned structures such as loops (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). When the MOSES dispersion direction happens to be aligned nearly perpendicular to the magnetic field, filamentary structures on the transition region serve almost as spectrograph slits unto themselves. The estimation of Doppler shifts then becomes a simple act of triangulation, and broadenings are also readily diagnosed (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018). A double-peaked profile can also be observed with sufficiently isolated features (T. Rust & C. C. Kankelborg 2019). Unfortunately, solar magnetic fields in the transition region are quite complex and do not have a global preferred direction. In cases where the field is nearly parallel to the instrument dispersion, spectral shifts and broadenings are not readily apparent.

The single diffraction grating also leads to a compromise in the optical performance of the instrument. Since the MOSES grating forms images in three orders simultaneously, there are not enough degrees of freedom in the optical system to control aberrations in all three spectral orders. During the first mission, MOSES was flown with a small amount of defocus (T. Rust & C. C. Kankelborg 2019), which exacerbated the inter-order point-spread function (PSF) variation and caused the individual PSFs to span several pixels (T. Rust & C. C. Kankelborg 2019;

S. Atwood & C. Kankelborg 2018). The outboard PSFs were elongated, with their major axes at different angles. This resulted in spurious spectral features that require additional consideration (S. Atwood & C. Kankelborg 2018) and further increase the complexity of the inversion process (T. Rust & C. C. Kankelborg 2019; H. T. Courrier & C. C. Kankelborg 2018).

As MOSES lacks a field stop, it is possible for the grating to image off-band radiation on the detector from solar features that lie outside the intended FOV. Although the intensity of this off-band radiation is relatively low, it is not negligible. J. D. Parker & C. C. Kankelborg (2022) compared synthetic MOSES images to the real data and found that approximately ten percent of the intensity in the zeroth order image originated from more than ten dim lines in the MOSES passband, most of them too faint to be visible in the dispersed images. This study revealed that an undispersed ($m = 0$) channel, although attractive due to its lack of spatial-spectral ambiguity, is of limited utility due to spectral contamination. Moreover, the FOV should be clearly defined, and the same, for each image, so that the spectral and spatial content of each image is unambiguous.

Finally, the exposure cadence of MOSES is hindered by a $\sim$6 s readout time for the CCDs (J. L. Fox 2011). The observing interval for a solar sounding rocket flight is typically about five minutes. Consequently, every second of observing time is precious, both to achieve adequate exposure time and to catch the full development of dynamical phenomena. The MOSES observing duty cycle is $\sim$50 % since it is limited by the readout time of its CCDs. Thus, valuable observing time is lost. The readout data gap compelled us to develop a sequence with exposures ranging from 0.25–24 s, a careful tradeoff between deep and fast exposures.

In summary, our experience leads us to conclude that the MOSES design has the following primary limitations:

1. inefficient use of volume
2. dispersion constrained by payload dimensions
3. too few dispersed images (orders)
4. single dispersion plane
5. lack of aberration control
6. poorly-defined and wavelength dependent FOV
7. spectral contamination
8. low duty cycle

In designing ESIS, we have sought to improve upon each of these points.

2.2. *ESIS Features*

The layout of ESIS (Figure 2) is a modified form of Gregorian telescope. Incoming light is brought to focus at an octagonal field stop by a parabolic primary mirror. In the ESIS layout, the secondary mirror of a typical Gregorian telescope is replaced by a segmented, octagonal array of diffraction gratings. From the field stop, the gratings re-image to CCD detectors arranged radially around the primary mirror. The gratings are optimized for first order, so that each CCD is fed by a single corresponding grating, and all the gratings are identical in design. The features of this new layout address all of the limitations described in Section 2.1, and are summarized here.

Replacing the secondary mirror with an array of concave diffraction gratings confers several advantages to ESIS over MOSES. First, the concavity of the gratings creates magnification in the ESIS optical system, which results in a shorter axial length than MOSES, without sacrificing spatial or spectral resolution. Second, the magnification and tilt of an individual grating controls the position of the dispersed image with respect to the optical axis, so that the spectral resolution is not constrained by the payload dimensions. Third, the radial symmetry of the design places the cameras closer together, resulting in a more compact instrument. Furthermore, by arranging the detectors around the optical axis, more dispersed grating orders can be populated; up to eight gratings can be arrayed around the ESIS primary mirror (up to six with the current optical table). This contrasts with the three image orders available in the planar symmetry of MOSES. Taken together, these three design features make ESIS more compact than MOSES (Section 2.1 item 1), improve spectral resolution (item 2) and allow the collection of more projections to better constrain the interpretation of the data (item 3).

The ESIS gratings are arranged in a segmented array, clocked in 45° increments, so that there are four distinct dispersion planes. This greatly aids in reconstructing spectral line profiles since the dispersion space of ESIS occupies a 3D volume rather than a 2D plane as with MOSES. For ESIS, there is always a dispersion plane within 22.5° of the normal to any loop-like feature in the solar atmosphere. As discussed in Section 2.1, a nearly perpendicular dispersion plane allows a filamentary structure to serve like a spectrographic slit, resulting in a clear presentation of the spectrum. This feature addresses Section 2.1 item 4.

Rather than forming images at three spectral orders from a single grating, each ESIS imaging channel has a dedicated grating. Aberrations are controlled by op-

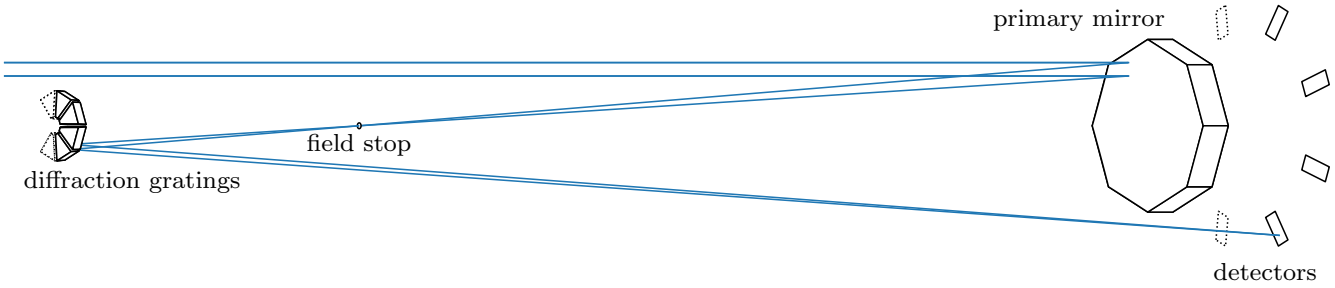


Figure 2. The ESIS optical layout. Dotted lines indicate the positions of unpopulated channels. The blue lines represent the path of O V through the system. The ESIS instrument is a pseudo-Gregorian design. The secondary mirror is replaced by a segmented array of concave diffraction gratings. The field stop at prime focus defines instrument spatial/spectral FOV. CCDs are arrayed around the primary mirror, each associated with a particular grating. Eight grating positions are available in principle; only six fit within the volume of the rocket payload. Four channels are populated for the first flight.

timizing the grating design to form images in first order, over a narrow range of ray deviation angles. This design controls aberration well enough to allow pixel-limited imaging, avoiding the PSF mismatch problems inherent to the MOSES design (Section 2.1 item 5). In its flight configuration with gratings optimized around a 629.732 Å wavelength, the instrument cannot be aligned and focused in visible light like MOSES. Visible gratings and a special alignment transfer procedure (Section 4.8) are used to align and focus ESIS.

The ESIS design also includes an octagonal field stop placed at prime focus. This confers two advantages. First, the field stop fully defines the instrument FOV, so that ESIS is not susceptible to the spectral confusion observed in MOSES data (Section 2.1 item 6). Second, each spectral line image observed by ESIS is bordered by the outline of the field stop (e.g., Figure 5). This aids the inversion process since outside of this sharp edge the intensity is zero for any look angle through an ESIS data cube. The size and octagonal shape of the field stop are defined by the requirement that all CCDs must see the entire FOV from edge to edge, while leaving a small margin for alignment.

Lastly, in contrast to MOSES, ESIS employs frame transfer CCDs to make optimum use of our five minutes of observing time. The ESIS design is shutterless, so that each detector is always integrating. The result is a 100 % duty cycle. The lack of downtime for readout also allows ESIS to operate at a fixed, rapid cadence of 10 s. Longer integration times can be achieved for faint features by exposure stacking (Section 2.1 item 8).

In summary, the ESIS concept addresses all the limitations of the MOSES design enumerated in Section 2.1. The volume of the ESIS optical layout is smaller than MOSES by almost a factor of two, yet with a smaller PSF, improved spectral resolution, and faster exposure cadence. ESIS offers several features to improve the recovery of spectral information, including more channels, crossed dispersion planes, and a field stop.

3. SCIENCE OBJECTIVES

Early flights of MOSES demonstrated a working concept of simultaneous EUV imaging and spectroscopy. This concept adds a unique capability to the science that we can obtain from the EUV solar atmosphere. ESIS as designed improves upon the MOSES concept, as discussed in the previous section, and therefore improves our ability to accomplish our scientific objectives. In this section, we set forth the specific scientific objectives of the ESIS mission. It is from these objectives that we derived the quantitative science requirements (Section 3.3) that drove the ESIS design.

The ESIS mission was designed to achieve the following two overarching science goals: (1) observe magnetic reconnection in the TR, and (2) map the transfer of energy through the TR with emphasis on magnetohydrodynamic (MHD) waves. These objectives have significant overlap with the missions of IRIS (B. De Pontieu et al. 2014), the EUV Imaging Spectrometer (EIS) (J. L. Culhane et al. 2007) aboard Hinode, the Extreme Ultraviolet Normal Incidence Spectrograph (EUNIS) (J. W. Brosius et al. 2007, 2014), and a long history of FUV and EUV slit spectrographs. The ESIS instrument, however, can obtain both spatial and spectral information co-temporally. This allows us to resolve complicated morphologies of compact TR reconnection events (as was done with MOSES (J. L. Fox 2011; T. L. Rust 2017; H. T. Courrier & C. C. Kankelborg 2018)) and observe signatures of MHD waves over a large portion of the solar disk. Therefore, in support of goal 1, we will use ESIS to map flows as a function of time and space in multiple TR reconnection events. To achieve goal 2, we will cross-correlate the evolution at multiple temperatures in the TR to map the vertical transport of energy over a wide FOV.

3.1. *Magnetic Reconnection Events*

Magnetic reconnection describes the re-arrangement of the magnetic topology wherein magnetic energy is converted to kinetic energy resulting in the acceleration of plasma particles. Reconnection is implicated in many dynamic, high-energy solar events. Solar flares are a well-studied example (e.g., E. R. Priest & T. G. Forbes 2002 and the references therein); however, we have little hope of pointing in the right place at the right time to observe a significant flare event in a rocket flight lasting only five minutes. Instead, we will search for signatures of magnetic reconnection in TR spectral lines. A particular signature of reconnection in the TR is the explosive energy release by ubiquitous, small-scale events. These explosive events (EEs) are characterized as spatially compact (≈1.5 Mm length (K. P. Dere 1994)) line broadenings on the order of 100 km s^{-1} (K. P. Dere et al. 1991). They are observed across a range of TR emission lines that span temperatures of 20 000–250 000 K (C II–O V) (D. Moses et al. 1994). The typical lifetime of an EE is 60–90 s (D. Moses et al. 1994; K. P. Dere 1994; K. P. Dere et al. 1991). Due to their location near quiet Sun magnetic network elements, and the presence of supersonic flows near the Alfvén speed, K. P. Dere et al. (1991) first suggested that EEs may result from the model of fast Petschek (H. E. Petschek 1964) reconnection.

The spectral line profile of EEs may indicate the type of reconnection that is occurring in the TR (e.g., T. L. Rust 2017). For example, the Petschek model of reconnection predicts a 'bi-directional jet' line profile with highly Doppler-shifted wings, but little emission from the line core (D. E. Innes & G. Tóth 1999). D. E. Innes et al. (2015) developed a reconnection model resulting from a plasmoid instability (A. Bhattacharjee et al. 2009). In contrast to the bi-directional jet, this modeled line profile has bright core emission and broad wings. Both types of profile are seen in slit spectrograph data (e.g., D. E. Innes et al. 1997, 2015, and the references therein); however, MOSES observed EEs with more complicated morphologies than either of these two models suggest (J. L. Fox et al. 2010; T. L. Rust 2017). It is unclear whether the differing observations are a function of wavelength and temperature, a result of a limited number of observations, or a consequence of the morphology of the event being difficult to ascertain from slit spectrograph data.

ESIS will observe magnetic reconnection in the context of EEs, by extending the technique pioneered by MOSES to additional TR lines. Explosive events are well suited to sounding rocket observations; a significant portion of their temporal evolution can be captured in >150 s (e.g., the analysis by T. L. Rust 2017) and they are sufficiently common to provide a statistically meaningful sample in a 5-minute rocket flight (e.g., K. P. Dere et al. 1989, 1991). In similarity with MOSES, we sought a TR line for ESIS that is bright and well enough isolated from neighboring emission lines so as to be easily distinguished.

3.2. *Energy Transfer*

Tracking the mass and energy flow through the solar atmosphere is a long-standing goal in solar physics. Bulk mass flow is evidenced by Doppler shifts or skewness in spectral lines. However, the observed non-thermal broadening of TR spectral lines may result from a variety of physical processes, including MHD waves (B. De Pontieu et al. 2015, 2007), high-speed evaporative upflows (e.g., nanoflares, S. Patsourakos & J. A. Klimchuk 2006), turbulence, and other sources (e.g., J. T. Mariska 1992). This is a broad topic which ESIS can address in many ways. Here we will focus on a single application; ESIS will search for sources of Alfvén waves in the solar atmosphere by observing Doppler shifts and line broadening as the spectroscopic signature of these waves. The spectroscopic signature of Alfvén waves is most obvious near the limb of the solar disk, since the motion of the ions is transverse with respect to the propagation direction. Pointing near the limb may make it more difficult to observe distinct EEs, described in Section 3.1, since the average width of TR lines often increases near the limb of the Sun (T. R. Ayres 2021), thus decreasing the signal-to-noise ratio (SNR) of single events.

Alfvén waves in coronal holes (CHs) are observed to carry an energy flux of 7×10^5 erg/cm^2/s, enough to energize the fast solar wind (M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). The source and frequency spectrum of these waves is unknown. Here, we hypothesize that MHD waves are similarly ubiquitous in quiet Sun (QS) and active regions (ARs), and play an important role in the energization of the quiescent corona.

The magnitude of non-thermal broadening of optically thin spectral lines is a direct measure of the wave amplitude (D. Banerjee et al. 2009; M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). We may estimate a lower limit on the non-thermal velocity to be observed as follows. We assume that the magnetic field is constant for small changes in scale height in the TR and that line-of-sight effects are negligible for observations sufficiently far from disk center. Since the solar wind is not accelerated to an appreciable fraction of the Alfvén wave velocity at altitudes $R \leq 1.15 R_\odot$ (S. R. Cranmer & A. A. van Ballegooijen 2005), the wave amplitude, v_{nt}, depends only weakly on electron density, n_e,

so that $v_{nt} \propto n_e^{-1/4}$ (M. Hahn & D. W. Savin 2013; T. G. Moran 2001). Assuming pressure balance between the low corona and TR, we may infer non-thermal velocities in the TR by scaling according to the temperature drop, $v_{nt} \propto T^{1/4}$. The measured non-thermal velocity of 24 km s^{-1} for Si VIII (J. G. Doyle et al. 1998) (0.8 MK (T. G. Moran 2003)) near the limb should, neglecting damping, correspond to velocities of at least 18 km s^{-1} in the O V (0.25 MK) line. This is larger than the thermal width of 11 km s^{-1}.

More recently, A. K. Srivastava et al. (2017) observed torsional Alfvén waves with amplitude ∼20 km s^{-1} and period ∼30 s in the chromosphere. Modeling shows that these torsional waves can transfer a significant amount of energy to the corona (T. Kudoh & K. Shibata 1999). The torsional motion is observed as Doppler shifts when viewed from the side. The oscillation period is long enough to be well resolved but short enough to see ∼10 cycles in a single rocket flight. An ESIS-like instrument is therefore well suited to observations of torsional Alfvén wave propagation over multiple heights in the TR.

By mapping Doppler velocities over a wide field of view in the TR, ESIS can address questions about both the origin of waves and whether they are able to propagate upward into the corona. Independent of the two propagation modes discussed above, there is a range of possible sources for Alfvén (and other MHD) waves in the solar atmosphere. Three potential scenarios are: (1) Waves originate in the chromosphere or below and propagate through the TR at a spatially uniform intensity; (2) Intense sources are localized in the TR, but fill only a fraction of the surface; and (3) Weak sources are localized in the TR, but cover the surface densely enough to appear like the first case. The resulting non-thermal widths for localized sources are significantly higher than the ∼18 km s^{-1} lower limit derived above. The concentration of non-thermal energy observed by ESIS will serve as an indicator of source density. Comparison of Doppler maps captured at different temperatures by ESIS will indicate whether a uniform source density originates in the chromosphere or below (scenario 1) or is associated with spatially distributed TR phenomena (scenario 3) such as explosive events, or macrospicules. Comparison with a wider selection of ground- and space-based imagery will allow us to determine whether intense, localized sources (scenario 2) are associated with converging or emerging magnetic bipoles, type II spicules, spicule bushes, or other sources beneath the TR. For these comparisons, we need only to localize, rather than resolve, wave sources. A spatial resolution of ∼2 Mm is sufficient to localize sources associated with magnetic flux tubes that are rooted in photospheric intergranular network lanes (e.g., T. E. Berger et al. 1995).

3.3. *Science Requirements*

ESIS will investigate two science targets: reconnection in explosive events and the transport of mass and energy through the transition region. The latter may take many forms, from MHD waves of various modes to EUV jets or macrospicules. To fulfill these goals, ESIS will obtain simultaneous intensity, Doppler shift and line width images of the O V 629.732 Å line in the solar transition region (0.25 MK) at rapid cadence. The bright, optically thin O V emission line is well isolated except for the two coronal Mg X lines (Figure 3). These coronal lines can be viewed as contamination or as a bonus; we expect that with the four ESIS projections it is possible to separate the O V emission from that of Mg X. From the important temporal, spatial, and velocity scales referenced in Sections 3.1 and 3.2 we define the instrument requirements in Table 1 that are needed to meet our science goals.

4. THE ESIS INSTRUMENT

ESIS is a multi-projection slitless spectrograph that obtains line intensities, Doppler shifts, and widths in a single snapshot over a 2D FOV. Starting from the notional instrument described in Section 2, ESIS has been designed to ensure all of the science requirements set forth in Table 1 are met. The final design parameters are summarized in Table 2.

A schematic diagram of a single ESIS channel is presented in Figure 4a, while the mechanical features of the primary mirror and gratings are detailed in Figures 4b and 4c, respectively.

4.1. *Optics*

Figure 4a shows the relative layout of the optics and detectors for a single ESIS channel. Here we give specific details of the primary mirror and gratings. The features of the field stop have been described previously in Section 2.2, while the CCD and cameras are covered in Section 4.10.

The primary mirror (Figure 4b) is octagonal in shape. The octagonal shape of the primary allows dynamic clearance for filter tubes that are arranged radially around the mirror (Section 4.6). The mirror is attached to a backing plate by three *bipods*: thin titanium structures that are flexible in the radial dimension, perpendicular to the mirror edge, but rigid in the other two dimensions. The bipods form a kinematic mount, isolating the primary mirror figure from mounting stress.

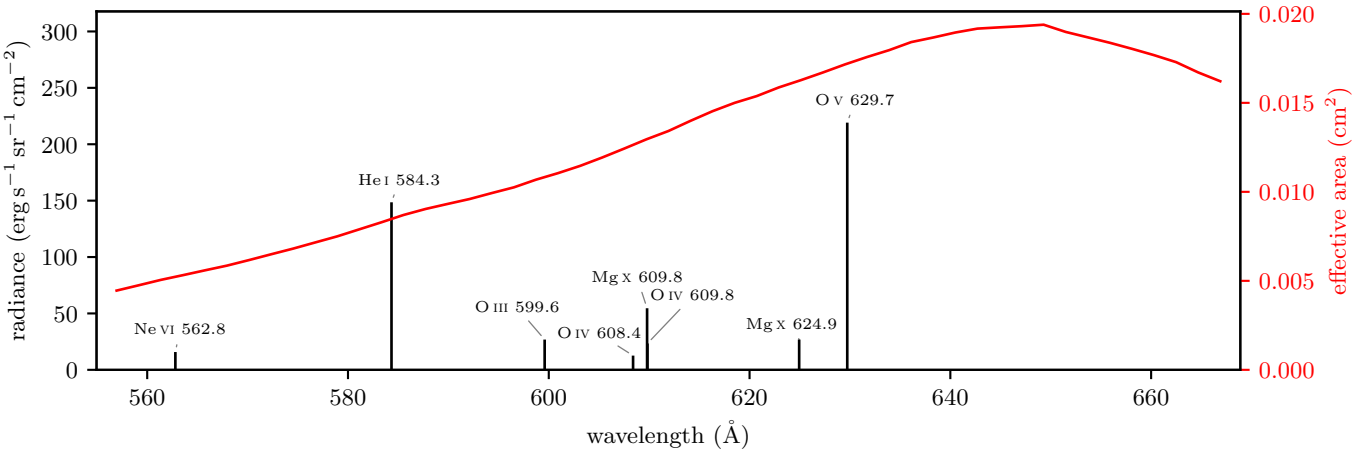


Figure 3. The eight brightest emission lines in the ESIS passband, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database (K. P. Dere et al. 1997; R. P. Dufresne et al. 2024), using the J. T. Schmelz et al. (2012) abundances of `sun_coronal_2012_schmelz_ext`, the quiet Sun differential emission measure (DEM) file `quiet_sun.dem`, and $n_eT = 1 \times 10^{15}$ K cm^{-3}.

Table 1. ESIS instrument requirements and capabilities. The capability for spatial resolution is the modulation transfer function (MTF) at the required resolution, which exceeds the MTF of 0.09 given by the Rayleigh criterion.

Parameter	Requirement	Science Driver	Capabilities
Spectral line	O V 629.732 Å	EEs	O V, Mg X, He I, Figure 3
Spectral sampling	18 km s^{-1}	Broadening from MHD waves	17.5 km s^{-1} pix^{-1}, Table 2
Spatial resolution	2.1″ (1.5 Mm)	EEs	MTF of 0.25, Table 4
SNR	17.3 (CH)	MHD waves in CH	17.6 (17 × 10 s exp.), Table 7
Cadence	15 s	Torsional waves	10 s eff., Section 4.7
Observing time	150 s	EEs	306.8 s, Section 5
FOV diameter	10′	Span QS, AR, and limb	11.5′, Table 2

The mirror had to maintain its figure under direct solar illumination, so a Corning ultra-low expansion (ULE) substrate was used. The transparency of ULE, in conjunction with the transparency of the mirror coating in visible and near-IR wavelengths (e.g., Table 2 and Section 4.6), helps minimize the heating of the mirror. Surface figure specifications for the ESIS optics are described in Section 4.2.

The spherical gratings (Figure 4c) re-image light from the field stop to form dispersed images at the CCDs. Each grating is individually mounted to a backing plate in a similar fashion as the primary mirror. For these much smaller optics, lightweight bipods were photochemically etched from thin titanium sheet. The bipods are bonded to both the grating and backing plate along the three long edges of each grating. The individual mounts allow each grating to be adjusted in tip and tilt to center the image on the CCD.

The gratings have a varied line space ruling pattern optimized to provide, in principle, pixel-limited imaging from the field stop to the CCDs. The pitch at the center of the grating is $d_0 = 0.38661$ μm (0.38677 μm as measured by the manufacturer) resulting in a dispersion of 17.5 km s^{-1} pix^{-1} at the center of the O V 629.732 Å FOV. Each grating serves only a single CCD. The groove profile is laminar, with depth optimized for $m = 1$. The modeled groove efficiency in this order is 40 %, close to the 41 % measured (Table 2).

Figure specification and groove profile are difficult to control near the edges of the gratings. Therefore, an uncoated margin was left around the edge of the grating, minimizing reflection in EUV (Figure 4c).

The ESIS passband is defined through a combination of the field stop, the grating dispersion, and the CCD size. The passband includes the He I 584.334 Å spectral line through Mg X (609.793 Å and 624.941 Å) to O V 629.732 Å. Figure 5 shows where images of each of the strong spectral lines fall on the CCD. The instrument dispersion satisfies the spectral resolution requirement in Table 1 and ensures that the spectral images

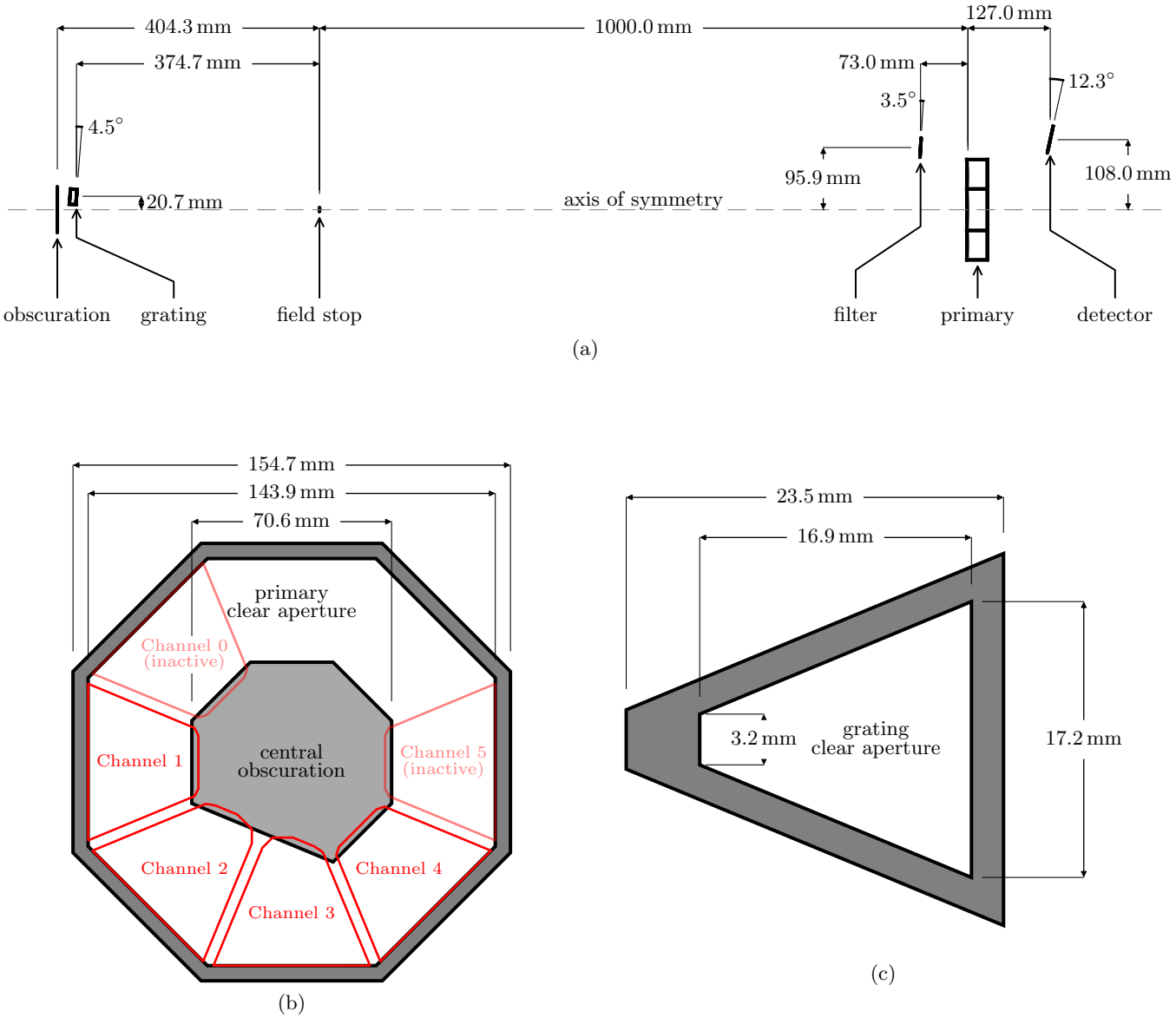


Figure 4. (a) Schematic diagram of a single channel of the ESIS optical system. (b) Clear aperture of the primary mirror, size of the central obscuration, and the footprint of the beam for each channel. (c) Clear aperture of one diffraction grating.

are well-separated; Figure 5 shows that He I 584.334 Å is completely separated from the target O V 629.732 Å line.

4.2. *Optimization and Tolerancing*

The science resolution requirement of 2.1″ (Table 1) was flowed down to specifications for the ESIS optics. To ensure that ESIS meets this requirement, an imaging error budget was developed to track parameters that significantly influence instrument resolution. The budget is roughly divided into two categories; the first includes 'variable' parameters that can be controlled to varying degrees (e.g., the figure and finish of the optics, grating radius and ruling, placement of the elements in the system, and the accuracy to which the instrument is focused). The second category consists of 'fixed' contributions (e.g., CCD charge diffusion, pointing stability, and diffraction from the entrance aperture). In this subsection we describe all these contributions to the instrument resolution budget.

Figure and surface roughness specifications for the primary mirror and gratings were developed first by a rule of thumb and then validated through a Fourier optics

Table 2. ESIS design parameters.

Component	Parameter	Calculated value (measured value)
Primary	Surface shape	Parabolic
	Focal length	1000.0 mm
	Aperture shape	Octagonal
	Aperture diameter	144 mm
	Coating	SiC single layer, optimized for O v 629.732 Å
	Efficiency (O v 629.732 Å)	25 % (19 %)
Field stop	Sky plane diameter	$11.5'$
	Aperture shape	Octagonal
	Aperture diameter	3.363 mm
Gratings (4)	Surface shape	Spherical
	Surface radius	597.830 mm (597.036 ± 0.156 mm)
	Aperture shape	Trapezoidal
	Aperture height	16.9 mm
	Aperture long base	17.2 mm
	Aperture short base	3.2 mm
	Ruling type	Varied line spacing
	Constant ruling spacing coefficient	0.38661 μm (0.38677 μm)
	Linear ruling spacing coefficient	-3.385×10^{-5} μm mm^{-1}
	Quadratic ruling spacing coefficient	-1.362×10^{-7} μm mm^{-2}
	Input angle	1.301°
	Output angle (O v 629.732 Å)	8.057°
	Manufacturing process	Individual master gratings
	Coating	Mg/SiC/Al multilayer, optimized for O v 629.732 Å
	Groove efficiency (O v 629.732 Å)	40 % (41 %)
	Efficiency (O v 629.732 Å)	14 % (13 %)
Filters (4)	Aperture shape	Circular
	Aperture diameter	30.0 mm
	Material	Al
	Thickness	100 nm
	Mesh transmission	82 %
	Mesh material	Ni
	Efficiency (O v 629.732 Å)	14 %
Detectors (4)	Manufacturer	Teledyne/e2v
	Model	CCD230-42
	Active area	2048 × 1024
	Pixel size	15 μm
	Quantum efficiency (O v 629.732 Å)	30 %
	Minimum cadence	1.2 s
System	Magnification	4.01
	f-number	$f/89$
	Plate scale along dispersion	$0.734''$ pix^{-1}
	Plate scale across dispersion	$0.770''$ pix^{-1}
	Dispersion	36.7 mÅ pix^{-1} (17.5 km s^{-1} pix^{-1})
	Passband	556.95 Å to 666.89 Å
	Efficiency (O v 629.732 Å)	0.10 %

model and Monte Carlo simulations. Surface figure errors were randomly generated, using a power law dis-

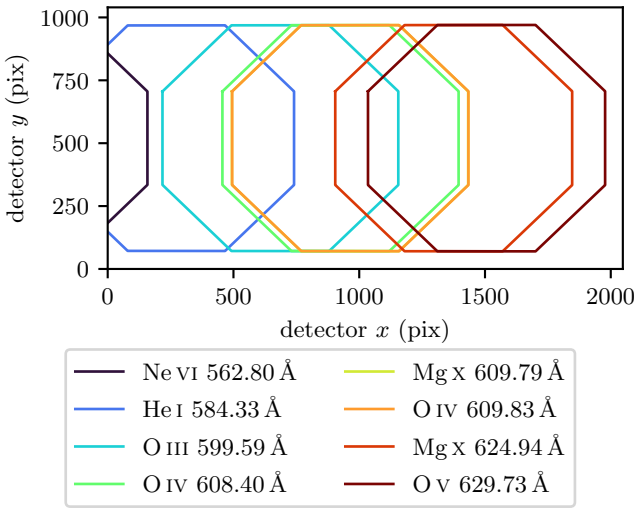


Figure 5. Areas occupied by strong spectral lines on the ESIS detectors. The plot axes are sized exactly to the CCD active area. The ESIS passband is defined by a combination of the field stop and grating dispersion.

tribution in frequency. The model explored a range of power spectral distributions for the surface figure errors, with power laws ranging from 0.1 to 4.0. For each randomly generated array of optical figure errors, the amplitude was adjusted to yield a target MTF degradation factor, as compared to the diffraction-limited MTF. For the primary mirror, the figure of merit was an MTF degradation of 0.7 at 2.1″ resolution. Though the grating is smaller and closer to the focal plane, it was allocated somewhat more significant MTF degradation of 0.6 based on manufacturing capabilities. The derived requirements are described in Table 3. Note that this modeling exercise was undertaken before the baffle designs were finalized. The estimated diffraction MTF and aberrations were therefore modeled for a rough estimate of the ESIS single sector aperture.

The initial grating radius of curvature, R_g, and ruling pattern of the ESIS gratings were derived from the analytical equations developed by L. Poletto & R. J. Thomas (2004) for stigmatic spectrometers. A second order polynomial describes the ruling pattern,

$$d = d_0 + d_1 x + d_2 x^2 \,, \tag{1}$$

where x is a Cartesian coordinate running outward from the optical axis with its origin at the center of the grating (Figure 4c). The parameters of Equation 1 and R_g were chosen so that the spatial and spectral focal curves intersect at the center of the O V 629.732 Å image on the CCD.

Starting from the analytically-derived optical prescription, a model of the system was developed in raytrace software. In the raytrace model, R_g, d_1, d_2, grating cant angle, CCD cant angle, and focus position were then optimized to minimize the RMS spot at select positions in the O V 629.732 Å FOV. The resulting performance is illustrated in Figure 6. Figure 7 maps the RMS spot radius across the FOV at the brightest lines in the ESIS passband, and Figure 8 shows how the RMS spot radius at the center of the FOV changes as the CCD is moved through focus. The optical prescription derived from the raytrace is listed in Table 2 and Figure 4.

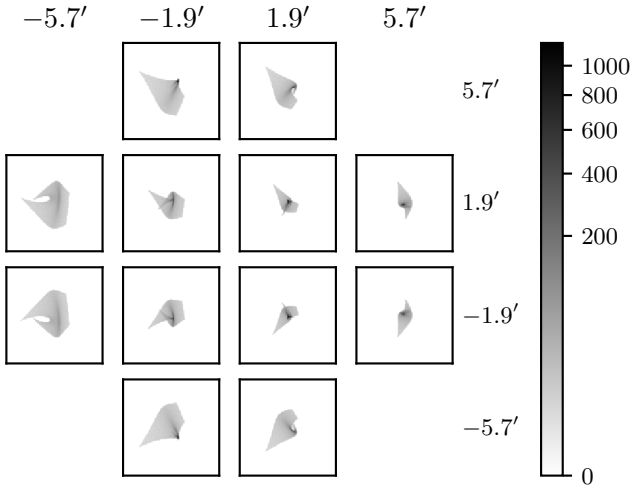


Figure 6. Raytraced spot diagrams for O V 629.732 Å with 4×4 field angles across the FOV. The box around each spot represents a single pixel on the detector. Each spot was traced using a stratified random grid across the pupil with 201×201 positions per spot.

The raytrace model was also used to quantify how mirror and positional tolerances affect the instrument's spatial resolution. Each element of the model was individually perturbed to either end of its tolerance, then a compensation was applied to adjust the image on the CCD. The compensation optimized the grating tip/tilt angles and focus position, so that the image was re-centered and the RMS spot radius minimized over a regular grid of points across the FOV. The defocus of the gratings and of the CCDs was not compensated, since each is a residual error of the focus itself. We then computed the change in the RMS spot radius at O V 629.732 Å, averaged over the FOV, between the optimized and perturbed models, and kept the larger change of the two ends of each tolerance.

The imaging error budget is displayed in Table 4. For the primary mirror and grating surface figure contributions, we list the MTF degradations calculated by the

Table 3. Figure and surface roughness requirements compared to metrology for the ESIS optics. Slope error (both the numerical estimates and the measurements) is worked out by first smoothing the surface with a boxcar as wide as the sample length, then dividing the difference in height across the integration length by that length.

Element	Parameter	Requirement	Measured
Primary	RMS slope error (μrad)	1.0	0.44
	Integration length = 4.0 mm		
	Sample length = 2.0 mm		
	RMS roughness (nm)	2.5	0.94
	Periods = 0.060–6.0 mm		
	RMS microroughness (nm)	1.0	0.55
	Periods = 1.6–70 μm		
Grating	RMS slope error (μrad)	3.0	1.9
	Integration length = 2.0 mm		
	Sample length = 1.0 mm		
	RMS roughness (nm)	4.0	1.6
	Periods = 0.024–2.4 mm		
	RMS microroughness (nm)	1.0	0.54
	Periods = 0.020–2.0 μm		

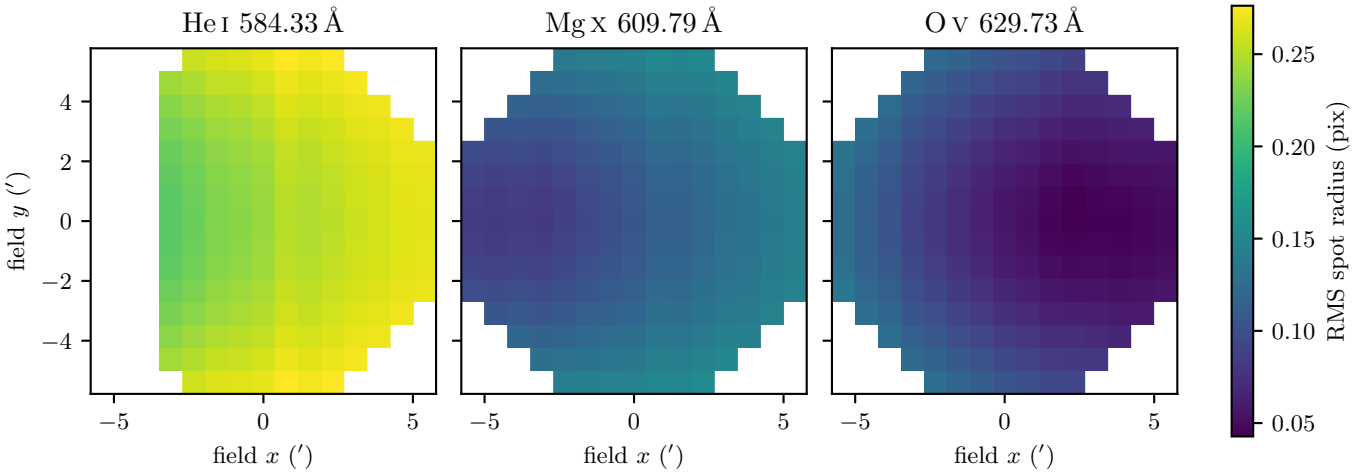


Figure 7. Maps of the RMS spot radius for the 3 brightest wavelengths in the ESIS passband. Each wavelength has 15×15 field points across the FOV, and each field point has a stratified random grid containing 15 × 15 pupil positions. The He I 584.334 Å line appears cropped since it is cut off by the edge of the detector.

Fourier optics code. To quantify the remaining entries, we assume that each term can be represented by a Gaussian function, of RMS radius σ, that "blurs" the final image. For each tolerance, $\sigma^2 = r^2 - r_0^2$, where r and r_0 are the RMS spot radii of the perturbed and optimized models found by the tolerance analysis described above, and the aberrations of the optimized model contribute $\sigma = r_0$ itself. Diffraction is approximated by a Gaussian fit to the diffraction pattern of a slit as wide as the beam on the primary mirror along the dispersion. Charge diffusion is computed with the method of J. R. Janesick (2001) for the model of the CCD described in Section 4.10. The pointing terms are the specifications of the Solar Pointing Attitude Rocket Control System (SPARCS) over an exposure of 10 s, with their peak-to-peak values converted to RMS as for a sinusoid, and the thermal drift of the instrument is allocated the same as the pointing drift. The value of the MTF in the right-most column of Table 4 is computed from each of the Gaussian blur terms at 0.48 cycles arcsec^{-1}, the spatial frequency of the 2.1″ resolution requirement. From Table 4, we estimate the total MTF of ESIS at its required 1.5 Mm resolution to be 0.25. By way of comparison, the widely used Rayleigh criterion (Lord Rayleigh

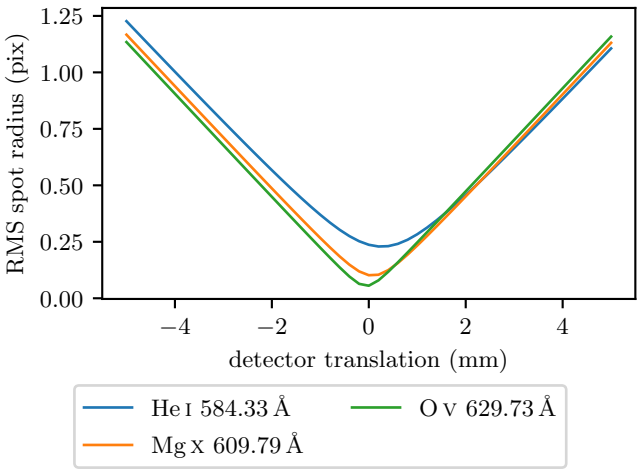


Figure 8. Focus curve for the field angle at the middle of the ESIS FOV for the 3 brightest wavelengths in the passband.

F.R.S. 1879) for diffraction from a circular aperture corresponds to an MTF of 0.09 at the resolution limit.

4.3. *Vignetting*

The original design of ESIS had no vignetting thanks to a stop placed at the primary mirror that was designed to perfectly fill the grating with the same amount of light for each point in the FOV. This is the ESIS design that was used for the optimization procedure of the grating parameters described in Section 4.2, for example. All other results described in the paper use the fully-open system. Before flight, we decided to remove the primary aperture stop to increase the sensitivity of the instrument at the expense of introducing vignetting to the ESIS FOV. This was acceptable since the vignetting was found to be a simple linear field as shown in Figure 9, and could be removed in the post-processing phase.

4.4. *Distortion*

Imaging from the prime focus to the detector introduces distortion. Since the four ESIS channels are arranged in four different orientations, the distortion complicates the comparison of the channels, and we must correct for it. The distortion is due to two factors: first, the tilt of the detector as needed to maintain good focus over the FOV (L. Poletto & R. J. Thomas 2004); second, the anamorphic magnification of the grating (F. Schweizer 1979). Together they make the plate scale finer along the dispersion than across it ($0.734''\,\mathrm{pix}^{-1}$ against $0.770''\,\mathrm{pix}^{-1}$, Table 2), so the image of the field stop is stretched along the dispersion compared with the field stop magnified by the ratio of the grating arms (Figure 10).

We describe where each point of the FOV lands on the detector with the quadratic polynomial

$$\begin{aligned}(x', y') = \mathbf{C} &+ \mathbf{C}_x x + \mathbf{C}_y y + \mathbf{C}_\lambda \lambda \\ &+ \mathbf{C}_{xx} x^2 + \mathbf{C}_{xy} xy + \mathbf{C}_{x\lambda} x\lambda \\ &+ \mathbf{C}_{yy} y^2 + \mathbf{C}_{y\lambda} y\lambda + \mathbf{C}_{\lambda\lambda} \lambda^2\end{aligned} \tag{2}$$

where (x, y) is the field angle, λ is the wavelength, and (x', y') is the position on the detector. The coefficients are fit to rays traced through the model of a single channel at the three target lines in the passband, and Table 5 lists them. A linear polynomial is not enough: it misplaces the field by as much as 2.4 pixels, while the quadratic polynomial is within 0.026 pixels of the raytrace wherever the field lands on the detector (Figure 11).

4.5. *Measured Distortion*

Every channel of ESIS forms a dispersed image of the same field, and recovering line intensities, Doppler shifts and widths from the four images requires knowing where each point of the sky lands on each detector to a fraction of a pixel, in every exposure. The as-built model of the optics, assembled from the measured figures of the components and their nominal placement, correlates with the flight images at only 0.36 to 0.42; the placement of the optics as flown differs from nominal by enough to move the images by several pixels. We therefore fit the model to the flight itself, and the fit is described in Appendix A. In brief, each channel's grating angles, ruling spacing, pointing and sensor placement were fit against the exposure at the middle of the flight, using the AIA 304 and 193 Å images of the same field as a proxy for the scene; the windows the field stop cuts into each image were placed on their measured edges, which separates the pointing from the gratings; the four channels were registered against one another on the sky at He I and O V; and the pointing, the drift of the windows and the focus of the primary mirror were followed exposure by exposure. Table 6 gives the correlation of each channel's model with its exposure after each stage, and over six exposures across the flight with each exposure's own pitch and yaw applied.

Figure 12 shows what moved during the flight. The pointing drifted by $6.2''$ in yaw and $1.4''$ in pitch, and the windows drifted from the reference exposure by up to 1.5 pix, which the model carries as a rotation of each channel's grating of up to $1.5''$. With those applied, the channels' skies still slid against one another by 0.26 pix rms, and by up to 0.52 pix: a translation of each channel's whole image inside its window, the same at both lines and smooth in time, which nothing after the field stop can produce. Each channel views the Sun through

Table 4. Imaging error budget and tolerance analysis results. MTF is given at 0.48 cycles arcsec^{-1}.

Element		Tolerance	σ (μm)	MTF
System	Aberrations		1.21	0.992
	Diffraction		3.82	0.922
	Thermal drift		1.01	0.994
Primary	Figure	Table 3		0.700
	Decenter x	±1.0 mm	0.57	0.998
	Decenter y	±1.0 mm	0.44	0.999
Grating	Figure	Table 3		0.600
	Radius	±2.4 mm	0.33	0.999
	Decenter x	±1.0 mm	0.51	0.999
	Decenter y	±1.0 mm	0.27	1.000
	Defocus	$\pm15\,\mu$m	0.90	0.996
	Clocking	±13 mrad	0.00	1.000
	Ruling density	$\pm1.0\,\mathrm{mm}^{-1}$	0.05	1.000
	Ruling d_1	$\pm5.1\times10^{-7}\,\mu\mathrm{m\,mm}^{-1}$	1.12	0.993
	Ruling d_2	$\pm8.6\times10^{-9}\,\mu\mathrm{m\,mm}^{-2}$	0.14	1.000
CCD	Decenter x	±1.0 mm	0.19	1.000
	Decenter y	±1.0 mm	0.09	1.000
	Defocus	±0.51 mm	1.78	0.983
	Charge diffusion		7.52	0.732
SPARCS	Pointing jitter	$\pm0.25''$	3.61	0.930
	Pointing drift	$0.014''\,\mathrm{s}^{-1}$	1.01	0.994
	Roll jitter	$\pm18''$	0.76	0.997
	Roll drift	$0.040^\circ\,\mathrm{h}^{-1}$	0.01	1.000
Total MTF				0.249

Table 5. Coefficients of the quadratic distortion model, Equation 2, of a single ESIS channel, fit at the three target lines in the passband. The field angles x and y are measured from the center of the FOV, and the wavelength λ from 607.95 Å, the mean of the three lines. The detector coordinates x' and y' are in pixels from the lower left corner of the light-sensitive area of the detector.

Coefficient		x'	y'
$\mathbf{C}$	(pix)	9.13×10^{2}	5.20×10^{2}
$\mathbf{C}_x$	(pix arcsec^{-1})	1.36	8.95×10^{-7}
$\mathbf{C}_y$	(pix arcsec^{-1})	-1.25×10^{-7}	1.30
$\mathbf{C}_\lambda$	(pix Å^{-1})	2.72×10^{1}	-7.79×10^{-6}
$\mathbf{C}_{xx}$	(pix arcsec^{-2})	6.92×10^{-6}	3.70×10^{-9}
$\mathbf{C}_{xy}$	(pix arcsec^{-2})	1.09×10^{-9}	4.40×10^{-6}
$\mathbf{C}_{x\lambda}$	(pix Å^{-1} arcsec^{-1})	2.52×10^{-4}	-4.10×10^{-8}
$\mathbf{C}_{yy}$	(pix arcsec^{-2})	2.07×10^{-6}	1.21×10^{-9}
$\mathbf{C}_{y\lambda}$	(pix Å^{-1} arcsec^{-1})	2.50×10^{-8}	9.37×10^{-5}
$\mathbf{C}_{\lambda\lambda}$	(pix Å^{-2})	2.45×10^{-3}	3.35×10^{-7}

its own sector of the primary mirror, and a change of focus moves its image along its own dispersion while the field stop's edges stay put; the motion is reproduced, to the precision of the measurement, by a focus that differs from sector to sector and drifts through the flight (Appendix A). The focus of the primary as a whole drifted by 31 μm; the sectors departed from that mean by up to 18 μm, channel 0's at the end of the flight, and channel 1's sector moved the most, by 56 μm; a paraboloid has one focus for every zone, so if this is real it is the figure of the mirror changing unevenly with temperature, by about 100 nm of sag between sectors. The edges the measurement rests on are soft on several sides and the two lines disagree on their motion by up to half the effect, so the data do not exclude that part of the sector pattern is the apparent position of those edges drifting; the model carries the simpler reading, one focus per sector, with that caveat. With it, over the 27 exposures bright enough to measure, the channels agree to 0.11 pix rms at He I and 0.08 pix at O V, and at worst 0.28 pix and 0.24 pix. What the sector focus leaves, 0.08 pix rms in the registration measurement itself, is also fit as an empirical offset of each channel's pointing, of up to 0.15″ (0.2 pix), which reduces it to 0.05 pix; the model

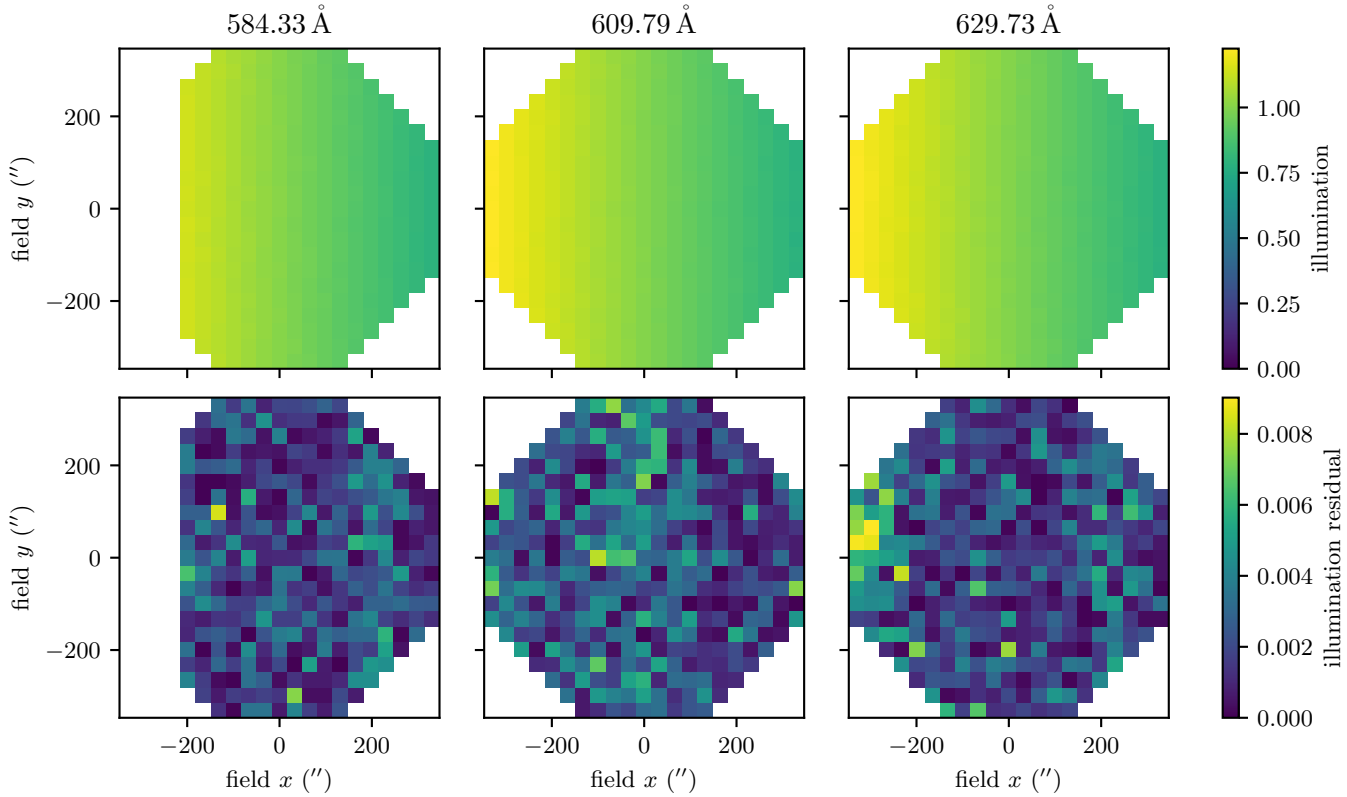


Figure 9. (Top) The relative illumination of a single ESIS channel as a function of position in the FOV, at each of the three target lines in the passband. The illumination is the fraction of the pupil which is unvignetted, normalized so that its average over the FOV is unity. (Bottom) The residual between that illumination and a linear model of it. Both rows are plotted only where the model was fit: the cells inside the FOV whose light lands on the detector. In the leftmost column, at He I, the FOV is cut off on the left by the edge of the detector.

carries these twelve numbers as an option, off by default, since no mechanism stands behind them (Appendix A). Only the component of a misregistration along a channel's dispersion is a velocity error, and a pixel along the dispersion is 18.9 km s^{-1} at He I and 17.4 km s^{-1} at O V: projected that way, the registration of the channels contributes 1.5 km s^{-1} and 0.6 km s^{-1} rms to a measured Doppler shift, and 4.1 km s^{-1} at worst, against quiet-Sun velocities of order 10 km s^{-1}. The three faintest exposures that close the flight show no window edges to measure, so the drift of their windows is extrapolated, and the registration is poorest there, reaching 0.9 pix at He I in the last.

4.6. *Coatings and Filters*

The diffraction gratings are coated with a multilayer optimized for a center wavelength of O V 629.732 Å, developed by a collaboration between Reflective X-ray Optics LLC and Lawrence Berkeley National Laboratory (LBNL). LBNL characterized a single, randomly selected grating after it was coated. Its reflectivity in the $m = 1$ order is constant over the instrument FOV, while the $m = 0$ order is almost completely suppressed (Figure 13), and its efficiency is nearly uniform across its surface, away from its edges (Figure 14). Figure 15 shows a schematic of the coating, which consists of four layer pairs of silicon carbide (SiC) and magnesium (Mg). The aluminum (Al) layers are deposited adjacent to each Mg layer to mitigate corrosion. As Charles mentioned, this doesn't make sense. Why can it go SiC to Mg, but not Mg to SiC?

The maximum reflectance for the coating alone in the nominal instrument passband is ∼35 % in the upper panel of Figure 16, measured from witness samples coated at the same time as the diffraction gratings. Combined with the predicted groove efficiency from Section 4.1 and, given the relatively shallow groove profile and near normal incidence angle, the total reflectivity in first order is ∼14 % at O V 629.732 Å. This is confirmed by the first order efficiency measured from a single ESIS grating in the lower panel of Figure 16.

Unlike EUV imagers (e.g., TRACE (B. N. Handy et al. 1999), AIA (J. R. Lemen et al. 2012), and the High-Resolution Coronal Imager (Hi-C) (K. Kobayashi et al. 2014)) the ESIS passband is defined by a combination of the field stop and grating (Section 4.1, Figure 5)

Table 6. The correlation of each channel's model image with its exposure after each stage of the distortion fit, against the reference exposure, and the mean over 6 exposures across the flight, the reference among them, with each exposure's own pitch and yaw applied. The last row is how far the internal alignment leaves each channel's sky from channel 1's at the reference exposure, in pixels.

Stage	Channel 0	Channel 1	Channel 2	Channel 3
Capture	0.811	0.848	0.802	0.828
Windows placed	0.810	0.848	0.802	0.828
Shared polish	0.815	0.848	0.804	0.828
Across the flight	0.811	0.845	0.802	0.824
Alignment residual [pix]	0.01	0.00	0.03	0.02

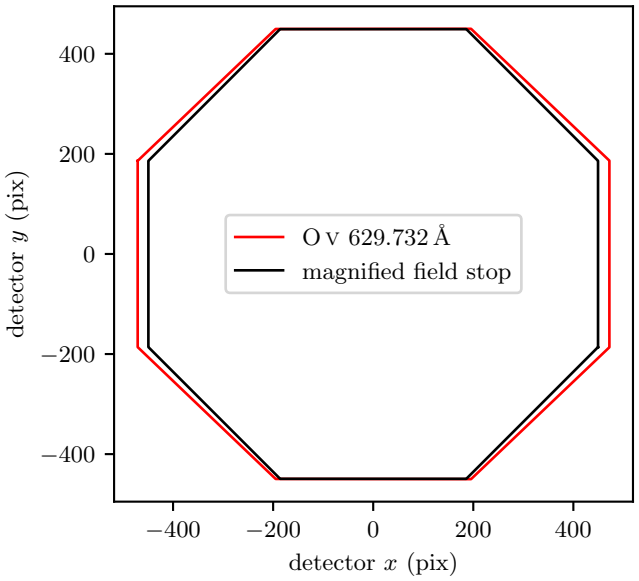


Figure 10. Plot of the magnified, undistorted field stop aperture vs. the distorted O v 629.732 Å image of the field stop aperture on the ESIS detector. The magnification factor used for the undistorted field stop aperture is the ratio of the grating exit arm to the grating entrance arm (4.01). The distorted image of the field stop aperture was calculated using the ESIS distortion model, described in Table 5.

rather than multilayer coatings. The coating selectivity is therefore not critical in this respect, allowing the multilayer to be manipulated to suppress out-of-band bright, nearby emission lines. The lower panel of Figure 16 shows the peak reflectance of the grating multilayer is shifted slightly towards longer wavelengths to attenuate the He I 584.334 Å emission line, reducing the likelihood of detector saturation. A similar issue arises with the bright He II 303.785 Å line. Through careful design of the grating multilayer, the reflectivity at this wavelength is ∼2.5 % of that at O v 629.732 Å (upper panel of Figure 16). In combination with the primary mirror coating (described below) the rejection ratio at 303.785 Å is ∼37 dB. Thus, He II 303.785 Å emission is completely attenuated at the CCD.

The flight and spare primary mirrors were coated with the same Al/SiC/Mg multilayer. Corrosion of this multilayer rendered both mirrors unusable. The failed coating was stripped from primary mirror SN001. The mirror was then re-coated with a 5 nm thick layer of chromium (Cr) to improve adhesion followed by a 25 nm thick layer of SiC. The reflectance of this coating deposited on a silicon (Si) wafer witness sample appears in Figure 16. The spare primary mirror (SN002) retains the corroded Al/SiC/Mg multilayer.

The Si CCDs are sensitive to visible light as well as EUV. Visible solar radiation is much stronger than EUV, and visible stray light can survive multiple scatterings while retaining enough intensity to contaminate the EUV images. Luxél (F. R. Powell et al. 1990) Al filters 100 nm thick were used to shield each CCD from visible light. The Al film is supported by a 70 line per inch (lpi) nickel (Ni) mesh, with 82 % transmission. The theoretical filter transmission curve, modeled from CXRO data (B. L. Henke et al. 1993), is displayed in Figure 16. We conservatively estimate filter oxidation at the time of launch as a 4 nm thick layer of Al_2O_3.

An Al filter is positioned in front of the focal plane of each CCD by a filter tube, creating a light-tight box with a labyrinthine evacuation vent (e.g., Figure 19). The placement of the filter relative to the CCD is optimized so that the filter mesh shadow is not visible. By modeling the filter mesh shadow, we find that a position far from the CCD (200 mm) and mesh grid clocking of 45° to the detector array reduces the shadow amplitude well below photon statistics. The MOSES instrument utilizes a similar design; no detectable signature of the filter mesh is found in data and inversion residuals from the 2006 MOSES flight.

To prevent oxidation, and to minimize the risk of tears, pinholes, and breakage from handling, the filters were stored in a nitrogen purged environment until after payload vibration testing.

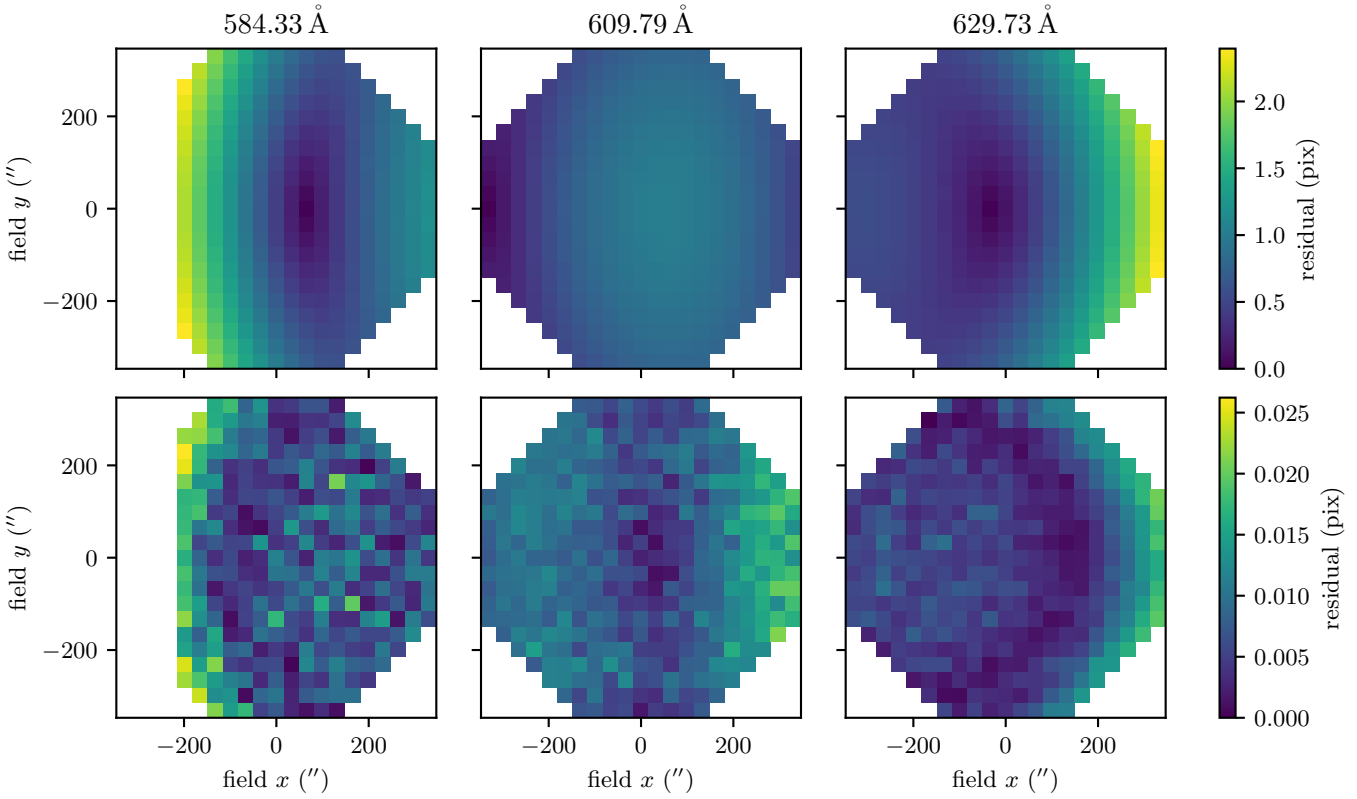


Figure 11. Magnitude of the residual between a linear distortion model and the raytrace model (top) and between a quadratic distortion model and the raytrace model (bottom). This figure demonstrates that a quadratic distortion model is sufficient to achieve sub-pixel accuracy.

4.7. *Sensitivity and Cadence*

Count rates for ESIS are estimated using the expected component throughput from Section 4.6 and the CCD quantum efficiency (QE) listed in Table 2. Line intensities are derived from J. E. Vernazza & E. M. Reeves (1978) (V&R) and the Solar and Heliospheric Observatory (SOHO)/Coronal Diagnostic Spectrometer (CDS) (R. A. Harrison et al. 1995) data, and are given in a variety of solar contexts: QS, CHs, and ARs. The 100 % duty cycle of ESIS (Section 4.10) gives us the flexibility to use the shortest exposures that are scientifically useful. So long as the shot noise dominates over read noise (which is true even for our coronal hole estimates at 10 s exposure length), we can stack exposures without a significant SNR penalty. Table 7 shows that ESIS is effectively shot noise limited with a 10 s exposure. The signal requirement in Table 1 is met by stacking exposures. Good quality images (∼300 counts) in active regions can be obtained by stacking 30 s worth of exposures. This cadence is sufficient to observe explosive events, but will not resolve torsional Alfvén waves described in Section 3.2. However, by stacking multiple 10 s exposures, sufficient SNR *and* temporal resolution of torsional Alfvén wave oscillations can be obtained.

Table 7. Estimated signal statistics per channel (in photon counts) for ESIS lines in CH, QS, and AR. Note that the SNR estimates are lower bounds since charge diffusion decreases the shot noise.

Source	V&R	V&R	V&R	CDS
Solar context	QS	CH	AR	AR
1 × 10 s exp.				
O v 629.732 Å	25	21	76	39
Mg x 624.941 Å	4	0	29	17
Total	29	22	105	56
Shot noise	5.4	4.6	10.2	7.5
Read noise	2.0	2.0	2.0	2.0
SNR	5.0	4.3	10.1	7.2
17 × 10 s exp.				
Total	488	366	1782	954
SNR	20.8	17.6	41.5	29.9

4.8. *Alignment and Focus*

In the conceptual phase of ESIS, the decision was made to perform focus and alignment in visible light with a helium-neon (He-Ne) source. Certain difficulties are introduced by this choice, however, the benefits out-

weigh the operational complexity and equipment that would be required for focus in EUV. Moreover, a sounding rocket instrument requires robust, adjustment-free mounts to survive the launch environment. Such a design is not amenable to iterative adjustment in vacuum. The choice of alignment wavelength is arbitrary for most components; CCD response and multilayer coating reflectivity is sufficient across a wide band of visible wavelengths. The exceptions are the thin film filters (which were not installed until just before launch and have no effect on telescope alignment and focus) and the diffraction gratings. Visible light gratings were manufactured specifically for alignment and focus. These gratings are identical to the EUV flight version, but with a ruling pattern scaled to a 632.8 nm wavelength.

After alignment and focus had been obtained with the He-Ne source, the instrument was prepared for flight by replacing the visible gratings by the EUV flight versions. Each grating (EUV and alignment) is individually mounted to a backing plate using a bipod system similar to that of the primary mirror. The array of gratings on their backing plates are in turn mounted to a single part which we call the 'tuffet.' The backing plate is attached to the tuffet by bolts through spherical washers. With this mounting scheme, the gratings can be individually aligned in tip/tilt. The tuffet is attached to the secondary mirror mount structure (Figure 17). This enables the entire grating array to be replaced simply by removing/installing the tuffet, such as when switching between alignment and EUV gratings.

Properly positioning the gratings was expected to be the most difficult aspect of final telescope assembly. Table 4 shows that the telescope is very sensitive to defocus. The depth of field is an order of magnitude smaller in EUV than in visible light. Moreover, the sensitivity to defocus at the gratings is $M^2 + 1 = 17$ times greater than at the detectors, where $M = 4.01$ is the ratio of the grating exit arm to the grating entrance arm. Another sensitive aspect of the telescope is grating tip/tilt. A tolerance of $\sim \pm 0.34$ mrad is needed to ensure that the entire image lands on the active area of the CCD.

Once the visible gratings were aligned and focused, the challenge was to transfer this alignment to the EUV flight gratings. We performed this transfer using a 4D Phasecam 6000 interferometer. We aligned the interferometer to each visible grating such that the tip, tilt, and defocus aberrations were zero. Then, placing the corresponding flight grating in the visible grating's place, we shimmed the mounting screws of the flight grating to match the tip, tilt, and defocus of the alignment grating. Since the mounting interface for the tuffet is extremely repeatable by design we were able to swap tuffets and compare their alignment and focus to ensure everything was transferred correctly.

When transferring the focus and alignment two key details were considered. First, there is nothing about the tuffet that constrains the grating roll. Therefore, we needed to ensure each flight grating had the same roll as each visible grating. This was accomplished by using a He-Ne laser diverged through a cylindrical optic that illuminated each grating with a line perpendicular better make sure I know which direction is implied here to the dispersion direction of the grating. The line of light was reflected back onto a ruled target that could be compared between gratings. Since our alignment gratings were ruled to image light of approximately an order of magnitude longer wavelength the laser and cylindrical optic were placed at the position of Littrow for the 10th order image in the visible grating, and the 1st order for the EUV gratings. Second, during testing we measured slight differences in radius of curvature between each grating. Therefore each flight grating was prescribed a specific amount of defocus to account for the difference in radius of curvature between each optic when transferring alignment and focus.

4.9. *Apertures and Baffles*

Needs a big rework Each channel of ESIS was designed with two apertures: one at the surface of the grating and a stop in front of the primary mirror. The purpose of the aperture at the grating is to mask the out-of-figure margins at the edges of these optics. This provides a well defined edge to the clear aperture of each grating while also keeping unwanted rays from being reflected from the grating margins and back onto the CCDs. The dimensions of the grating aperture match those of the grating clear aperture shown in Figure 4c.

The stop at the primary mirror was designed to be the stop of each channel. Its area was maximized under the constraint that no rays be vignetted anywhere else in the system. The gratings and their clear apertures were the most significant areas of concern for potential vignetting, so the shape of the stop was largely influenced by the shape of the grating clear aperture. The inner extent of the stop was defined by the occultation of the primary by the shadow cast from the gratings and their mounts. This presented an intricate geometry problem, as the occultation is a function of the incoming field angle, the radial extent of the grating mount, and the distance of the mount to the primary mirror along the optical axis. Hence, the inner extent of the stop was solved for iteratively with the optimization described in Section 4.2, which affected the placement of the gratings relative to the primary mirror.

After final optimization, the stop geometry was analyzed with the optical model to check for vignetting at the grating. A footprint diagram was generated at the grating from multiple grids of rays. The incidence angle of each grid of rays corresponded to the extremes of the FOV defined by the positions of the eight points of the octagonal field stop. The footprint diagram showed that, with the stop completely filled, no ray landed outside of the grating clear aperture, and no ray was intercepted by the central obscuration.

This non-vignetting stop left considerable surface area of the primary mirror unused, which could only be put to use by relaxing the vignetting constraint. Before flight, the stop was removed to increase the sensitivity of the instrument at the expense of vignetting (Section 4.3), which made the grating aperture the stop of each channel. The footprint of the beam of each channel on the primary mirror in Figure 4b is that of the instrument as flown, without the stop.

The ESIS baffles are designed to block direct light paths between the front aperture plate and the CCDs for any ray $<1.4°$ from the optical axis. This angle is purposefully larger than the angular diameter of the Sun ($\sim 0.5°$) so that any direct paths are excluded from bright sources in the solar corona. All baffles are bead-blasted, anodized Al sheet metal oriented perpendicular to the optical axis. The size and shape of the cutouts were determined using a combination of the raytrace from Section 4.2 and 3D modeling. Between the primary mirror and the gratings, each cutout is the footprint of the rays of that trace where they cross the baffle. Between the gratings and the CCDs, the light path is defined as the volume that connects each vertex of the grating clear aperture to every vertex of the image of the octagonal field stop on the CCD. This is a conservative definition that ensures no rays within the FOV are excluded, and therefore unintentionally vignetted by the baffles. The cutouts there are the projection of this volume onto the baffle. A conservative 1 mm margin is added to each cutout to prevent unintentional vignetting. A model of the six baffles, showing their cutouts and positions on the optical bench, is displayed in Figure 18.

4.10. *Cameras*

The ESIS CCD cameras were designed and constructed by MSFC and were the latest in a series of camera systems developed specifically for use on solar space flight instruments. The ESIS camera heritage includes those flown on both the Chromospheric Lyman-alpha Spectro-polarimeter (CLASP) (R. Kano et al. 2012; K. Kobayashi et al. 2012) and Hi-C (K. Kobayashi et al. 2014).

The ESIS detectors are CCD230-42 astro-process CCDs from Teledyne/e2v. For each camera, the CCD is operated in a split frame transfer mode with each of the four ports read out by a 16-bit A/D converter. The central 2048×1024 pixels of the $2k \times 2k$ device are used for imaging, while the outer two regions are used for storage. Two overscan columns on either side of the imaging area and eight extra rows in each storage region were intended to monitor read noise and dark current. When the camera receives the trigger signal, it transfers the image from the imaging region to the storage regions and starts image readout. The digitized data are sent to the data acquisition and control system (DACS) through a SpaceWire interface immediately, one line at a time. The frame transfer takes 50 ms, and readout takes 1.1 s. The cadence is adjustable from 1.2 s to 600 s in increments of 100 ms, to satisfy the requirement listed in Table 1. Because the imaging region is continuously illuminated, the action of frame transfer (transferring the image from the imaging region to the storage regions) also starts the next exposure without delay. Thus the exposure time is controlled by the time period between triggers. Camera 1 (Figure 19) generates the sync trigger, which is fed back into Camera 1's trigger input and provides independently buffered triggers to the remaining three cameras. The trigger signals are synchronized to better than ± 1 ms. Shutterless operation allows ESIS to observe with a 100 % duty cycle. The cadence is limited only by the 1.1 s readout time.

MSFC custom designed the camera board, enclosure, and mounting structure for ESIS to fit the unique packaging requirements of this experiment (Figure 19). The front part of the camera is a metal block which equalizes the temperature across the CCD while fastening it in place. The carriers of all cameras are connected to a central two-piece copper (3 kg) and aluminum (1 kg) thermal reservoir (cold block) by flexible copper cold straps. The flexible cold straps allow individual cameras to be translated parallel to the optical axis (by means of shims) up to $\sim$13 mm to adjust focus in each channel prior to launch. The centrally located cold block was cooled by liquid nitrogen (LN_2) flow from outside the payload until just before launch. The LN_2 flow was controlled automatically by a ground support equipment (GSE) computer so that all cameras were maintained above survival temperature but below the target temperature of $-55°$C to ensure a negligible dark current level.

The gain, read noise, and dark current of the four cameras were measured at MSFC using an ^{55}Fe radioac-

tive source. Cameras are labeled 1, 2, 3, and 4 with associated serial numbers SN6, SN7, SN9, and SN10 respectively in Figure 19. Gain ranges from 2.50 to 2.63 $e^- DN^{-1}$ in each quadrant of all four cameras. Table 8 lists gain, read noise, and dark current by quadrant for each camera.

The QE of the ESIS CCDs was not measured before flight. This paper models it with the measurements of a back-illuminated e2v CCD97 by I. Moody et al. (2017) and J. Heymes et al. (2020), which give a QE of 39 % at He I 584.334 Å and 30 % at O V 629.732 Å.

Table 8. ESIS camera properties.

Camera	Quad.	Gain ($e^- DN^{-1}$)	Read noise (e^-)	Dark current ($e^- s^{-1}$)
1 (SN6)	1	2.57	10.3	0.061
	2	2.50	10.5	0.051
	3	2.52	10.7	0.049
	4	2.53	9.7	0.061
2 (SN7)	1	2.55	10.4	0.117
	2	2.58	10.8	0.102
	3	2.57	10.7	0.156
	4	2.63	10.8	0.175
3 (SN9)	1	2.57	10.6	0.059
	2	2.53	10.4	0.050
	3	2.52	10.4	0.054
	4	2.59	11.1	0.067
4 (SN10)	1	2.60	10.1	0.117
	2	2.60	10.1	0.100
	3	2.54	10.7	0.125
	4	2.58	10.5	0.153

4.11. *Avionics*

The ESIS DACS is based on the designs used for both CLASP (R. Kano et al. 2012; K. Kobayashi et al. 2012; N. Narukage et al. 2016) and Hi-C (K. Kobayashi et al. 2014; L. A. Rachmeler et al. 2019). The electronics are a combination of military off-the-shelf (MOTS) hardware and custom designed components. The DACS is a 6-slot, 3U, open VPX PCIe architecture conduction cooled system using an AiTech C873 single board computer. The data system also includes a MOTS PCIe switch card, MSFC parallel interface card, and two MOTS SpaceWire cards. A slot for an additional SpaceWire card is included to accommodate two more cameras for the next ESIS flight. The C873 has a 2.4 GHz Intel i7 processor with 16 GB of memory. The operating temperature range for the data system is −40 °C to 85 °C. The operating system for the flight data system is Linux Fedora 23.

The DACS is responsible for several functions; it controls the ESIS experiment, responds to timers and uplinks, acquires and stores image data from the cameras, downlinks a subset of images through telemetry, and provides experiment health and status. The DACS is housed with the rest of the avionics (power supply, analog signal conditioning system) in a 0.56 m to 0.43 m transition section outside of the experiment section. This relaxes the thermal and cleanliness constraints placed on the avionics. Custom DC/DC converters are used for secondary voltages required by other electronic components. The use of custom designed converters allowed additional ripple filtering for low noise.

4.12. *Pointing System*

The imaging target was selected prior to launch, the morning of the day of flight. During flight, pointing was maintained by the SPARCS (Lockheed Missiles and Space Co. 1969). Images from Camera 1 were downlinked and displayed in real time on the SPARCS control system console at intervals of ∼16 s to verify pointing was maintained during flight.

4.13. *Mechanical*

ESIS and MOSES are mounted on opposite sides of a composite optical table structure originally developed for the Solar Plasma Diagnostic Experiment (SPDE) (M. E. Bruner 1995). The layered carbon fiber structure features a convenient, precisely coplanar array of threaded inserts with precision counterbores. The carbon fiber layup is designed to minimize the longitudinal coefficient of thermal expansion. The optical table is housed in two 0.56 m diameter skin sections, with a total length of 3 m. A ball joint and spindle assembly on one end and flexible metal aperture plate on the other hold the optical table in position inside the skin sections. The kinematic mounting system isolates the optical table from bending or twisting strain of the skins.

5. MISSION PROFILE

ESIS was launched on a Terrier Black Brant sounding rocket from White Sands Missile Range on 2019 September 30, alongside MOSES. J. D. Parker et al. (2022) describe the flight, the conditions on the Sun, and the data in detail, so we summarize the flight only briefly here. The cameras exposed continuously from launch, at a cadence of 10 s, and the exposures taken while the payload shutter door was closed serve as darks. The door opened 73.0 s after launch. The ring-laser gyroscope of the SPARCS held the pointing from 124.2 s to 431.0 s

after launch, which gives the observing time of 306.8 s in Table 1, and the door closed at 433.0 s. The Sun was very quiet that day, so ESIS was pointed at disk center, and thermal expansion inside the instrument made the image drift slowly across the detectors during the flight (J. D. Parker et al. 2022). The images were stored on board and retrieved after the payload descended on a parachute and was recovered, while some were also sent to the ground during the flight (Section 4.12). The mechanical shutter of MOSES failed, so no MOSES data were obtained during this flight.

6. CONCLUSIONS AND OUTLOOK

ESIS is a next generation slitless spectrograph, designed to obtain co-temporal spectral and spatial images of the solar transition region and corona. In this paper, we present details of the scientific objectives, instrument, image and spectral resolution, data acquisition, and flight profile.

Operating at a 10 s cadence, ESIS acquired spectra over its full FOV ∼30 times during its sounding rocket flight.

ESIS follows on the proven MOSES design, incorporating several design changes to improve the utility of the instrument. The symmetrical arrangement of CCDs and diffraction gratings results in a compact instrument while increasing the number of dispersed images and dispersion planes. This aids the inversion process, while also allowing access to higher order spectral line profile moments. Individual gratings improve resolution by controlling aberration in each channel. The addition of a field stop eliminates spectral contamination and provides an easily recognizable edge for data inversion. The ESIS design also demonstrates that all this can be accomplished in a volume small enough to serve as a prototype for a future orbital instrument.

For the first flight, four of the six available ESIS channels were populated with optics optimized around the O V emission line. The large (11.5′), high resolution FOV (1.54″, 73 mÅ) can simultaneously observe the evolution of small scale EUV flows and large scale MHD waves in high temporal cadence. ESIS also enables the study of transport of mass and energy in the transition region and corona during the ∼5 minute data collection portion of rocket flight.

ESIS was recovered after a successful first launch on 2019 September 30, with first results from its data reported by J. D. Parker et al. (2022). A second flight, ESIS-II, is planned for 2027, with new optics for the Ne VII and Si XII lines, and the instrument will be refined further with an eye toward orbital opportunities. Suborbital flights such as ESIS-II allow us to expand the instrument to its full complement of six channels and refine our data analysis methods, but do not provide access to major flares and eruptive events that drive space weather. The long term prospect is that an ESIS-like instrument on an orbital platform could provide high cadence maps of spectral line profiles in solar flares, allowing unique and comprehensive observations of the dynamics in solar eruptive events, flare ribbons, and the flare reconnection region.

ACKNOWLEDGMENTS

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Software: aastex (R. T. Smart 2026a), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), euv-snapshot-imaging-spectrograph (R. T. Smart et al. 2026a), fiasco (Will Barnes et al. 2026), Matplotlib (J. D. Hunter 2007), msfc-ccd (R. T. Smart & J. D. Parker 2026a), named-arrays (R. T. Smart & J. D. Parker 2026b), NumPy (C. R. Harris et al. 2020), optika (R. T. Smart et al. 2026b), utu (R. T. Smart 2026b)

APPENDIX

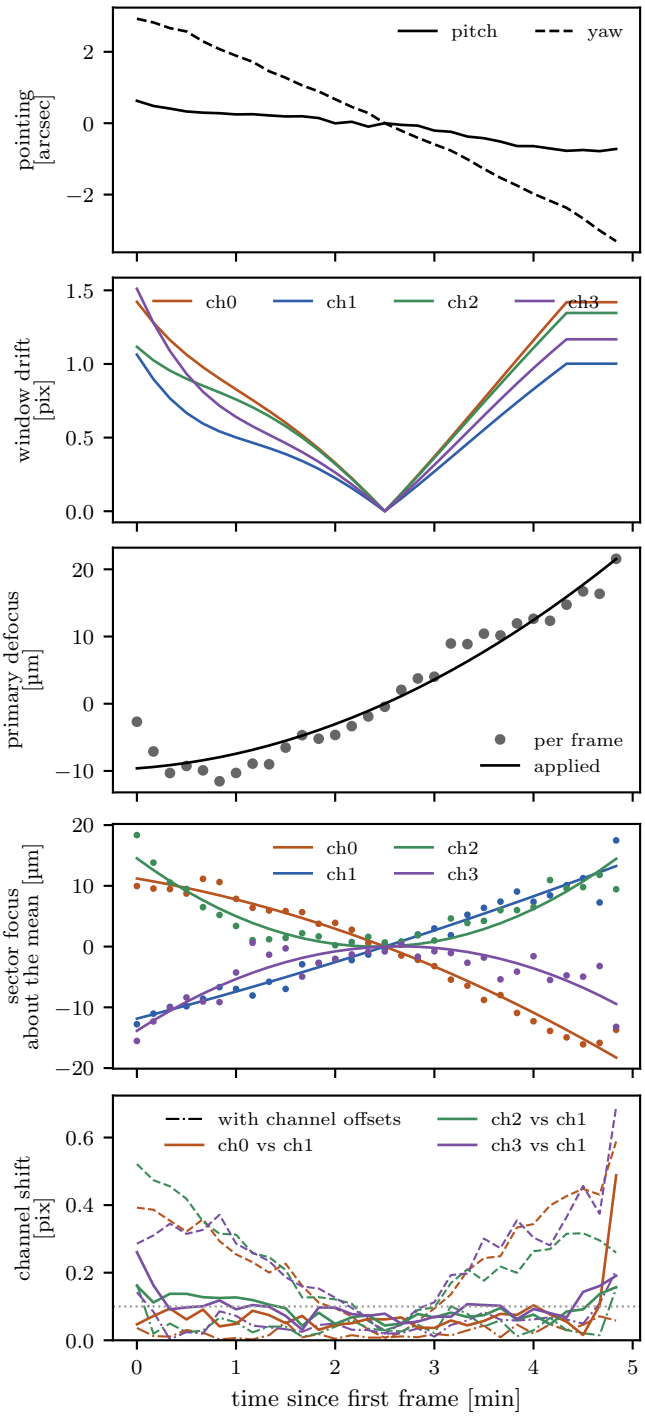


Figure 12. The distortion fit through the flight. From the top: the pointing of the payload fit to each exposure against the AIA scene; the drift of each channel's windows measured from the edges of the field stop; the defocus of the primary mirror at the field stop, the mean over its sectors, measured from the channels against one another in every exposure and smoothed by a quadratic; the focus of each channel's sector of the primary about that mean, measured per exposure (dots) and as applied (lines); and the shift of each channel's sky against channel 1, dashed with the focus held at its value at the reference exposure, solid with the focus of each sector following the flight and dash-dotted with the optional channel offsets as well, with the 0.1 pix acceptance threshold dotted.

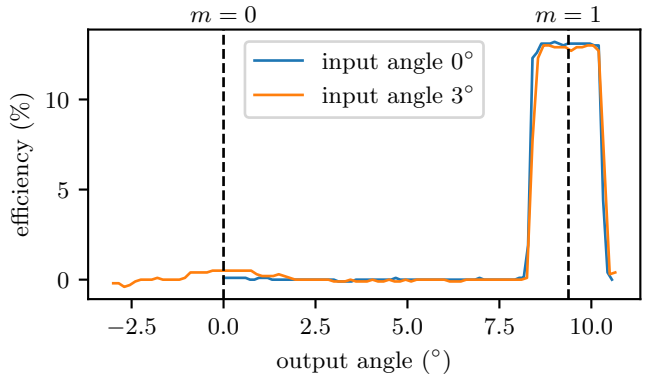


Figure 13. Measured efficiency at 63 nm of the Channel 2 grating as a function of reflection angle on 2018 January 21. Note flat response in first order over instrument FOV and suppression of zero order.

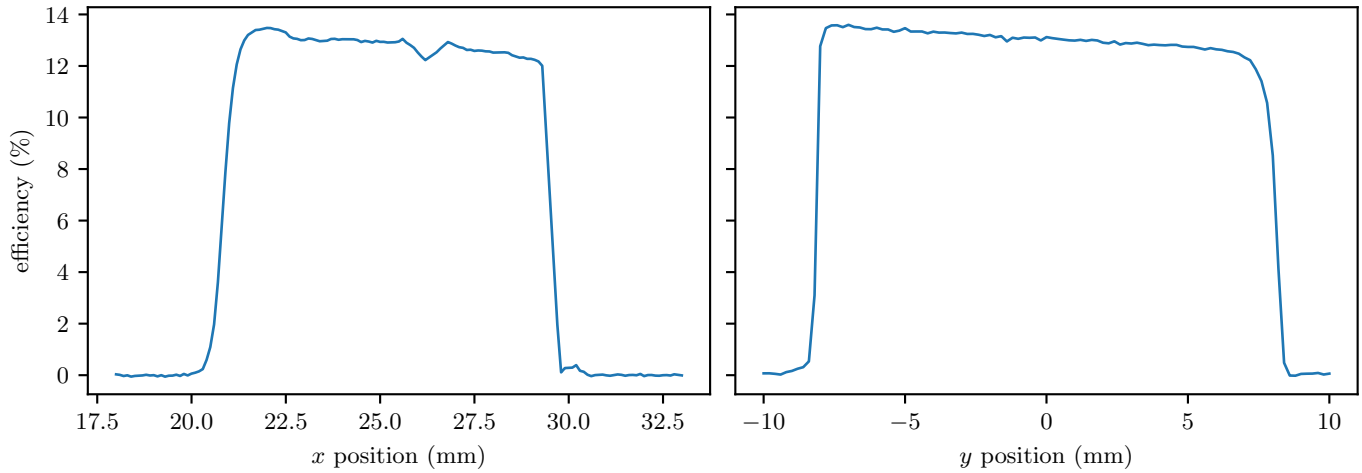


Figure 14. Channel 2 grating efficiency at 63 nm vs. position for two orthogonal slices across the optical surface on 2018 January 21.

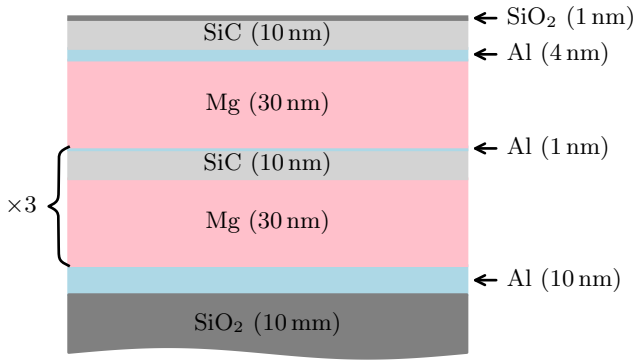


Figure 15. Schematic of the Al/SiC/Mg multilayer, with $N = 4$ pairs of SiC and Mg layers.

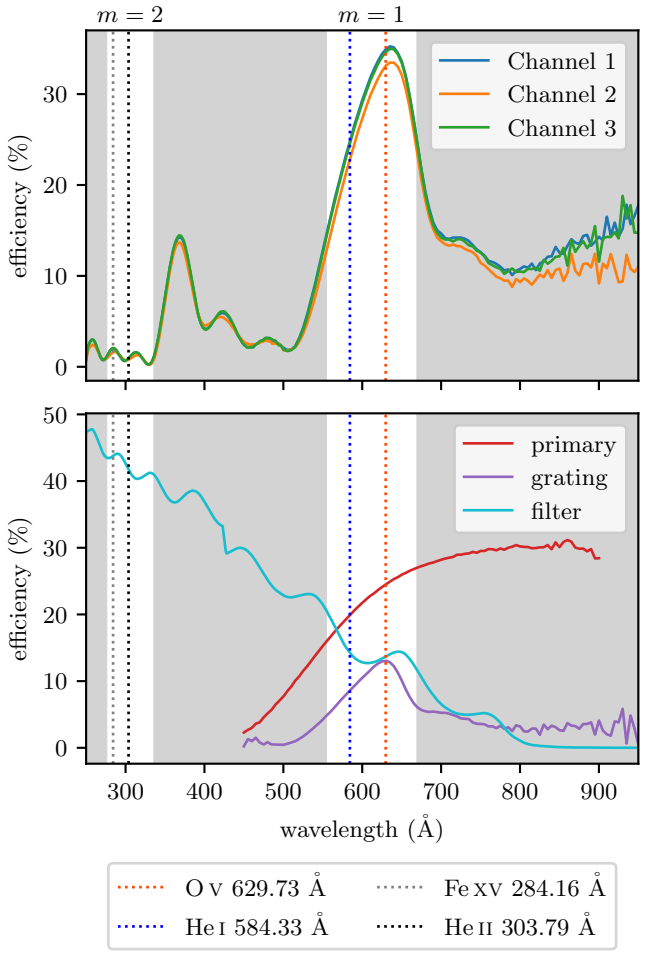


Figure 16. (Top) Measured reflectance for several multilayer coated witness samples at an incidence angle of 4° on 2018 January 21. The white regions indicate wavelengths that intercept the detector and the gray regions indicate wavelengths that miss the detector. Note the suppression of second order relative to the first order and the consistency of the coatings between each channel. The Channel 4 grating measurement is missing due to issues in the measurement apparatus. (Bottom) Comparison of the efficiency of the three main ESIS optical components: primary mirror, grating and filter. The primary mirror efficiency is based on measurements of a Si witness sample taken on 2018 May 10 at an angle of incidence of 4°. The grating efficiency is from a measurement of the Channel 2 grating taken on 2018 January 21 at an angle of incidence of 0°. The filter efficiency is a theoretical model that includes the filter mesh, 100 nm of Al and 4 nm of Al oxide.

A. THE DISTORTION FIT

The distortion of each channel is fit to the flight data as the placement of the optics in the as-built model, rather than as a polynomial over the detector, so that the result has physical units, carries the wavelength dependence the model knows, and can be checked against what was measured on the ground. The fit is reproducible from the `esis` package, whose documentation carries the full report, every table it commits, and the scripts that regenerate them.

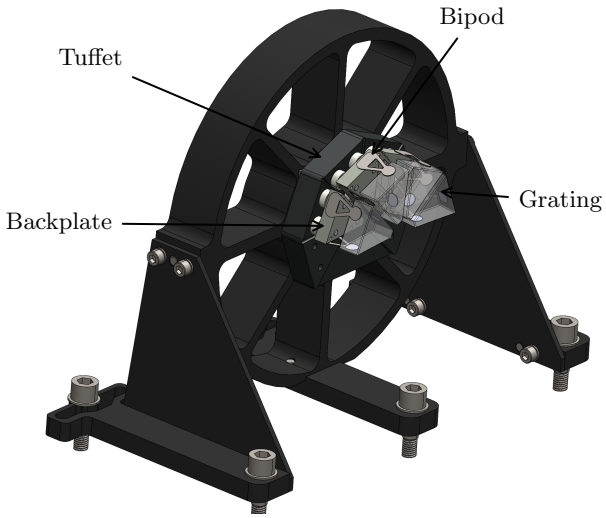


Figure 17. A solid model of the ESIS secondary mount. Each trapezoidal grating is supported and attached to its backplate by three bonded, titanium bipods. All mounted gratings sit on the "tuffet" and are mounted with three spherical washers. This allows independent tip/tilt adjustment of each grating, and for all of them to be removed/reinstalled without compromising alignment and focus.

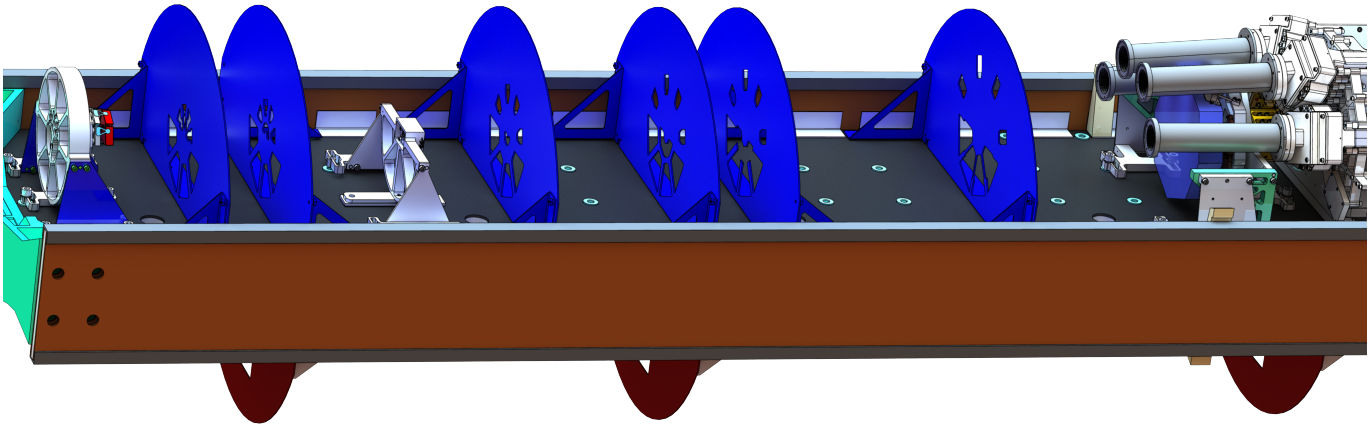


Figure 18. Model view of ESIS baffle placement and cutouts.

What one exposure determines—A single exposure of one channel determines eight combinations of that channel's parameters: a singular-value decomposition of the Jacobian of the sky-to-detector mapping with respect to the parameters has eight singular values above 40 pixels of image motion per half-width of the parameter bounds, then a cliff. Nine terms span them: the yaw and pitch of the grating, the ruling spacing, the pitch and yaw of the pointing, and the sensor's distance along the beam and its three angles. The pointing and the grating angles are degenerate in the image of the sky, which only the field stop breaks: its octagonal image, the window of each line on the detector, does not follow the pointing.

Merit—Every candidate model is linearized at the three brightest lines of the passband, He I, Mg X and O V, into a polynomial mapping from the sky to the detector and a vignetting function, and the AIA images of the same field at the time of the exposure, 304 Å for He I and O V and 193 Å for Mg X, are resampled through it onto the detector. The merit is the correlation of that model image with the exposure, which is indifferent to the photometric calibration

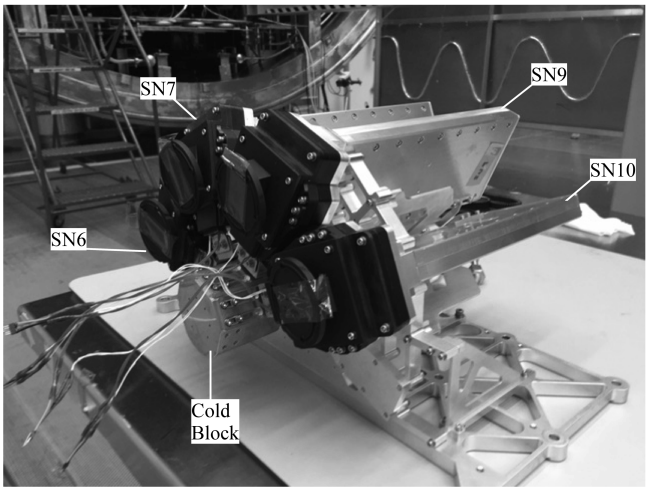


Figure 19. ESIS camera assembly as built by Marshall Space Flight Center (MSFC). Thin film filters and filter tubes are not installed in this image.

and to the smooth gradients vignetting and effective area leave between the channels. One evaluation takes about a second.

Stages—

1. *Capture.* Each channel is fit alone against the exposure at the middle of the flight, a seeded differential evolution over the nine terms followed by a restarted Nelder–Mead polish.

2. *Edges.* An error-function step is fit where each window's predicted outline crosses the rows and columns of every exposure bright enough to show it; the median over the flight places the windows and the departure from it in each exposure is the drift of the windows.

3. *Outline.* Each channel's windows are placed on the median edges through the grating angles and the sensor roll, which move the windows without moving the sky inside them, and the sky terms are re-polished.

4. *Shared.* The pointing is set to one value for the four channels, and each channel's grating and sensor placement is polished again with it fixed. The displacement of the primary is held at nominal, since a scan of it does not discriminate on the flight data.

5. *Internal alignment.* Each channel's exposure is read onto a common sky grid through its own mapping at He I and O V, inside the window that line illuminates, and cross-correlated tile by tile against channel 1; the parameters are moved so that the mappings reproduce the measured shifts. Two lines separate a geometric error, the same shift in both, from a dispersion error.

6. *Focus.* The channels are measured against channel 1 in every exposure, with the windows placed by the smoothed drift since the reference exposure, at which the reference was fit with no grating offset. Each channel views the Sun through its own sector of the primary, so a defocus moves its image along its own dispersion while the windows stay put; the focus of every sector is solved from the shifts through the model, four unknowns from six measured components, and smoothed by a quadratic through the flight. The mean over the sectors is the defocus of the primary as a whole.

7. *Pointing.* The pitch and yaw of every exposure are polished on the mean merit of the four channels, with the roll held, each channel's grating offset by the smoothed drift of its windows since the reference exposure, and each sector of the primary focused by that exposure's value.

8. *Acceptance.* The model is scored on six exposures across the flight, the reference exposure among them, with each exposure's own pitch and yaw applied, and in every exposure the coalignment metric is measured: the

length of the median tile shift of each channel against channel 1 at each line, its component along the channel's dispersion in pixels and in $\mathrm{km\,s^{-1}}$, the scatter of the tiles about it, and the decomposition of the tiles into the linear distortion modes with a test of whether what remains is noise (Figure 20).

The fit is stochastic: the capture draws its population at random, and every linearization samples the field and the pupil at random positions inside its cells. One seed sets all of these draws, and every committed table records the seed it was made with, zero. To see how much the draws matter, the whole chain was run from five seeds. The registration of the channels against one another, the median tile shift of each channel against channel 1 at each line and exposure, varies between the runs by 0.02 pix rms over the first 29 exposures, small against the registration itself. The scatter grows as the signal falls late in the flight, and reaches 0.08 pix rms in the last and faintest exposure, which holds 14 % of the peak signal. The parameters differ more, along the near-null directions of the problem, with the mapping differing by about 0.4 pixels in the median over the field and by up to about a pixel at its edge. In an earlier set of runs, on the code before its last corrections, one of the twenty captures settled in a false optimum, a correlation of 0.65 where the others reached 0.79 to 0.85, from which the later stages could not continue; that seed was not repeated and another stands in for it, and all twenty captures of this set reach 0.79 to 0.85. On the cluster's GPUs a seed repeats the fit only to about 0.0005 in correlation, not exactly.

The focus of each sector—With the pointing and the drift applied and the focus held at its value at the reference exposure, each channel's image still translated against the others' by up to 0.52 pix, rigid, the same at both lines and smooth in time. Tracking the interior of each window and its edges separately against the reference exposure, on the detector, the image did not move with the edges, so it was not a motion of a grating or a camera, which carry both; nor a decenter of the field stop or of the primary, which move every window, or every image, by one vector; and an error of plate scale or roll would have to be eight percent or five degrees. A focus that differs from sector to sector moves each channel's image along its own dispersion by its own amount, four parameters per exposure against six measured components, and it fits: it leaves the floor of the measurement, where the figure modes tried first, astigmatism, coma and trefoil, which force the sectors into a fixed relation, left three times that. Each sector's focus is smooth in time to the precision of a single exposure, a few microns. Nothing after the field stop can do this, and neither can any rigid motion of the mirror or the stop, so if it is real it is the figure of the primary changing unevenly with temperature, about 100 nm of sag between sectors over the flight, the same scale as the common drift of the focus. But the window edges, which the measurement takes for a rigid outline, are not one. The He I window has no left edge on any detector, where the image runs off the sensor, and two channels lack an edge in the other direction, so several windows are placed along an axis by a single edge. The edges differ in sharpness from 0.7 to over 4 pixels, several sharpen or soften by a pixel during the flight, their profiles are skewed, and the two lines, which share one field stop and one grating, disagree on the motion of the same side by up to 0.3 pixels over half the flight. The focus of a sector is measured along the channel's dispersion, and the window's position along the dispersion comes from the two edges perpendicular to it, which are the edges that behaved worst; the crossing of a skewed step moves when its width changes, by a fraction of the change, and that would appear here as a focus of the sector. The disagreement between the lines puts that at up to half the effect. The data therefore support two readings, the mirror bending or the along-dispersion fiducials drifting, and do not decide between them; the model carries the first, the simpler physical account, and the tables record the sectors' histories so that a thermal model of the mirror, or the next flight's temperature sensors, can judge it.

The optional channel offsets—With the focus of each sector applied, the channels' skies still differ from channel 1's by 0.08 pix rms over the bright exposures, close to the 0.05 pixels a single measurement resolves. The same measurement, repeated with the sector focus in place, fits what is left as an offset of each channel's own pitch and yaw, a quadratic in the exposure index that vanishes at the reference exposure and sums to zero over the channels so that the payload's pointing stays where the scene put it: twelve numbers, at most $0.15''$ or 0.2 pix on the sensor, which leave 0.05 pix rms. They are committed beside the fit and the flight model applies them only when asked, since they are empirical where the sector focus is a mechanism; an inversion can be run with and without them to see what a few hundredths of a pixel of registration do to the result.

Lessons—The field stop's edge was the calibration source, and its weak point: a field stop imaged whole, with margin on every side of every window, with edges verified sharp on the ground and a deliberate fiducial in each side, would have decided the question above and would fix the rotation and the scale of every window as well as its position. An

undispersed channel sharing the field stop would anchor the sky absolutely to a few hundredths of a pixel, where the AIA proxy anchors it to half a pixel, and would separate a common Doppler shift from a change of focus outright. The primary's focus drifted by 31 μm and the windows by up to 1.5 pix across the detectors, and each moved the images by tenths of a pixel; an athermal metering structure between the optics, or a way to measure both in flight, is worth more to the registration of the channels than any improvement of the detectors.

Cost—The capture takes one to two and a half hours per channel on one GPU, the later stages about three hours more, and the chain is about six hours of wall time on a cluster if no job is interrupted; on earlier code the same chain on the host without a GPU took about as long, and the stages from the focus on fit a workstation with 128 GB of memory.

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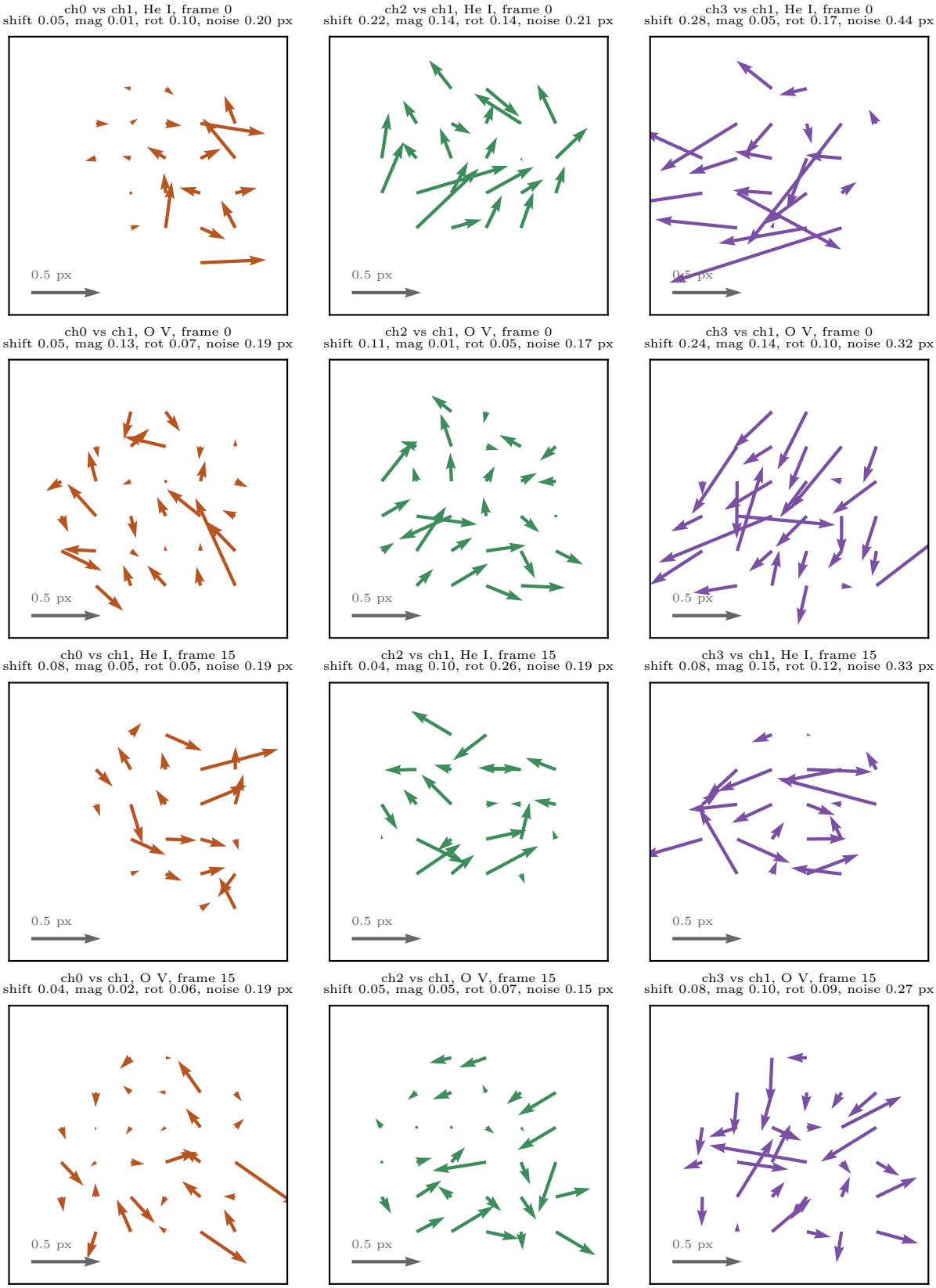


Figure 20. The coalignment of the channels over the field, after the fit. Each panel shows one channel's sky against channel 1's at one exposure and one line: the field is divided into tiles, each tile of the two channels is cross-correlated, and the shift that best registers them is drawn as an arrow at the tile's place in the field, with a half-pixel arrow for scale. The title of each panel decomposes the arrows into a translation, a magnification and a rotation of the field, and gives what remains once those are removed, which is uncorrelated from tile to tile and is the floor of the measurement. The first exposure of the flight is on top and the reference exposure, against which the fit was made, below.

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